MANUAL TESTING REPORT

Room Allocation Module • Hostel Management System
BCSE301P — Software Engineering Assessment 8

Report date	26 October 2026
Total test cases	65
Execution field	All test cases marked Executed (as requested)
Evidence package	87 screenshots supplied in ZIP

Execution/evidence note: The supplied ZIP manifest describes its images as script-captured screenshots of the **expected state**, not evidence of human execution. The cases are marked Executed to match the requested college-project format, but live behavior and pass/fail outcomes have not been independently verified. Do not interpret these screenshots as proof that each test was run against a live application.

1. Objective and scope

This report organizes the supplied manual test cases for the Room Allocation Module, including authentication, authorization, room/student workflows, allocation behavior, interface checks, and other scenarios represented in the source test-case file.

2. Execution status convention

Executed means the execution field has been marked Y for the requested report format. The Status column is recorded as “Executed”; it is not a Pass verdict. A verified Pass/Fail result requires running the steps in the target application and recording the observed result.

3. Test case details and screenshots

Each case below includes its test inputs and expected result, plus all screenshots associated with that test case in the supplied ZIP manifest where available.

1. AUTH-01 — Valid admin login

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P1 / UI only

Module: Authentication

Requirement: RA-11

Technique: UI only

Priority: P1

Scenario: Valid admin login

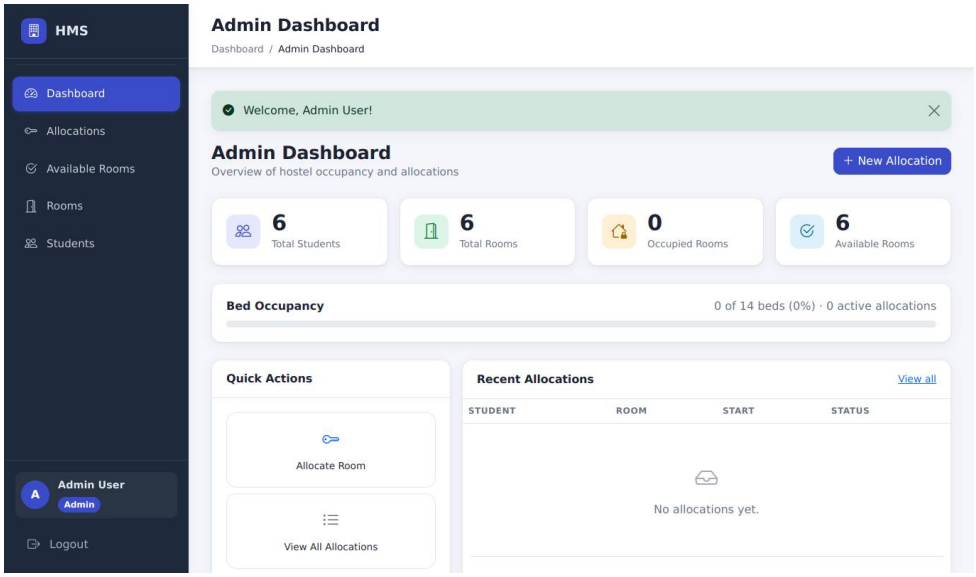
Preconditions: Server started with DEMO data (see TESTING_PROCEDURE.md section 3). Not logged in.

Steps: 1. Open http://127.0.0.1:5000/auth/login
2. Enter the admin e-mail and password
3. Click Login

Input Data: admin@hms.test / admin123

Expected Result: Redirected to the Admin Dashboard; message "Welcome, Admin User!" is shown; sidebar shows the admin menu.

AUTH-01_1.jpg — Admin dashboard after a successful login, with the welcome message
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

2. AUTH-02 — Valid warden login

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P2 / UI only

Module: Authentication

Requirement: RA-11

Technique: UI only

Priority: P2

Scenario: Valid warden login

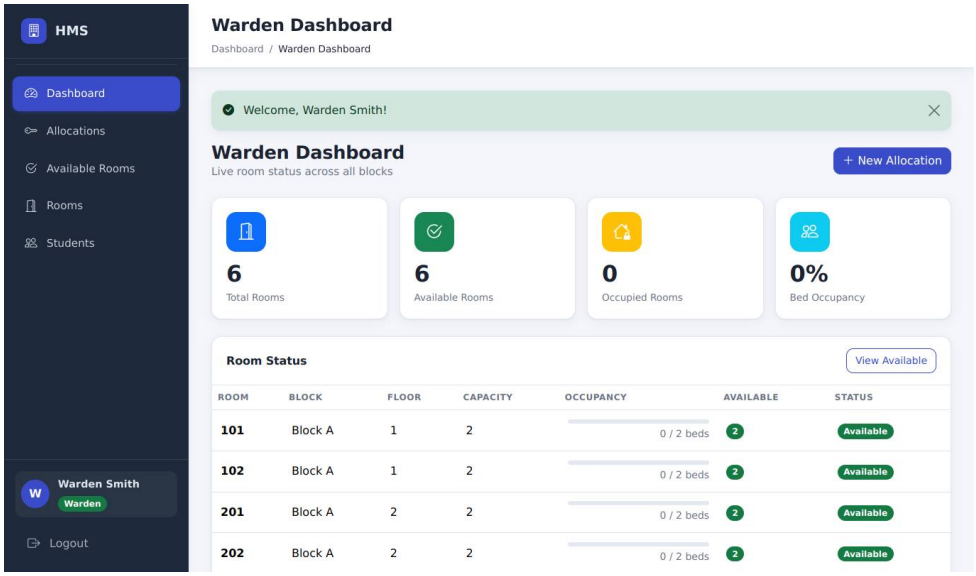
Preconditions: Not logged in.

Steps: 1. Open the login page
2. Enter the warden e-mail and password
3. Click Login

Input Data: warden@hms.test / warden123

Expected Result: Redirected to the Warden Dashboard; message "Welcome, Warden Smith!" is shown.

AUTH-02_1.jpg — Warden dashboard after login
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

3. AUTH-03 — Valid student login

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P2 / UI only

Module: Authentication

Requirement: RA-11

Technique: UI only

Priority: P2

Scenario: Valid student login

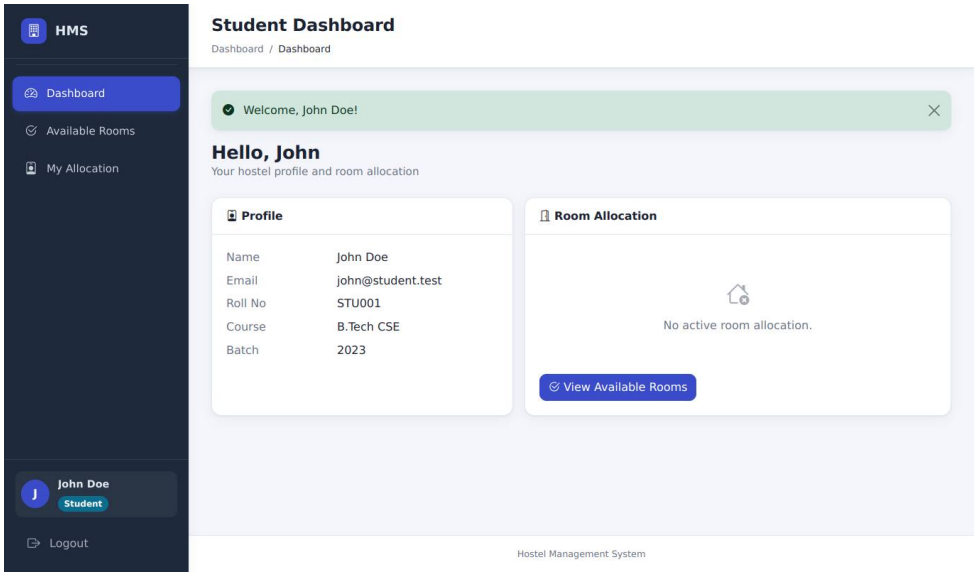
Preconditions: Not logged in.

Steps: 1. Open the login page
2. Enter the student e-mail and password
3. Click Login

Input Data: john@student.test / student123

Expected Result: Redirected to the Student Dashboard; message "Welcome, John Doe!" is shown.

AUTH-03_1.jpg — Student dashboard after login
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

4. AUTH-04 — Wrong password rejected

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P1 / UI only

Module: Authentication

Requirement: RA-11

Technique: UI only

Priority: P1

Scenario: Wrong password rejected

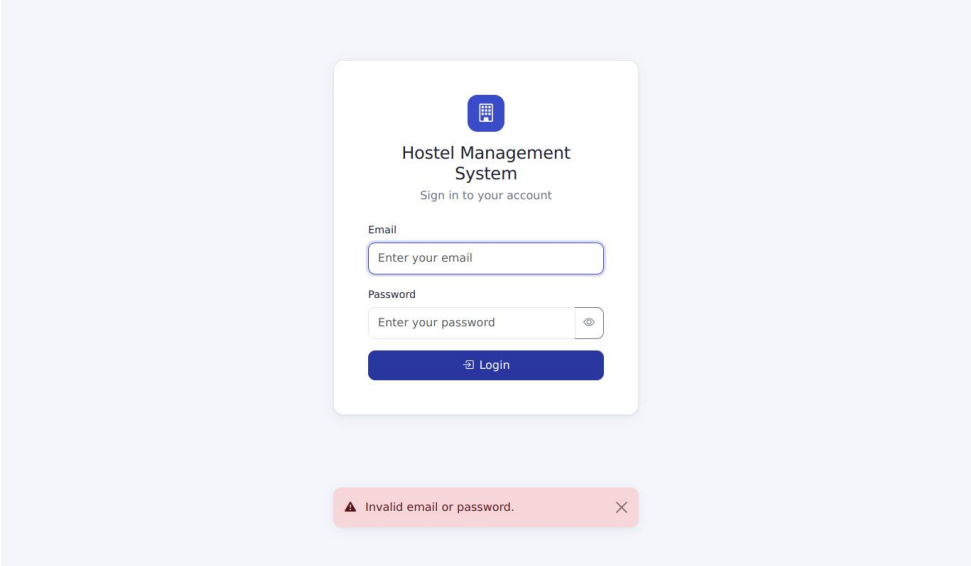
Preconditions: Not logged in.

Steps: 1. Open the login page
2. Enter admin@hms.test with password "wrong-password"
3. Click Login

Input Data: admin@hms.test / wrong-password

Expected Result: Stay on the login page; error "Invalid email or password." is shown; no session is created.

AUTH-04_1.jpg — Wrong password: error message, still on the login page
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

5. AUTH-05 — Unknown user rejected

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P2 / UI only

Module: Authentication

Requirement: RA-11

Technique: UI only

Priority: P2

Scenario: Unknown user rejected

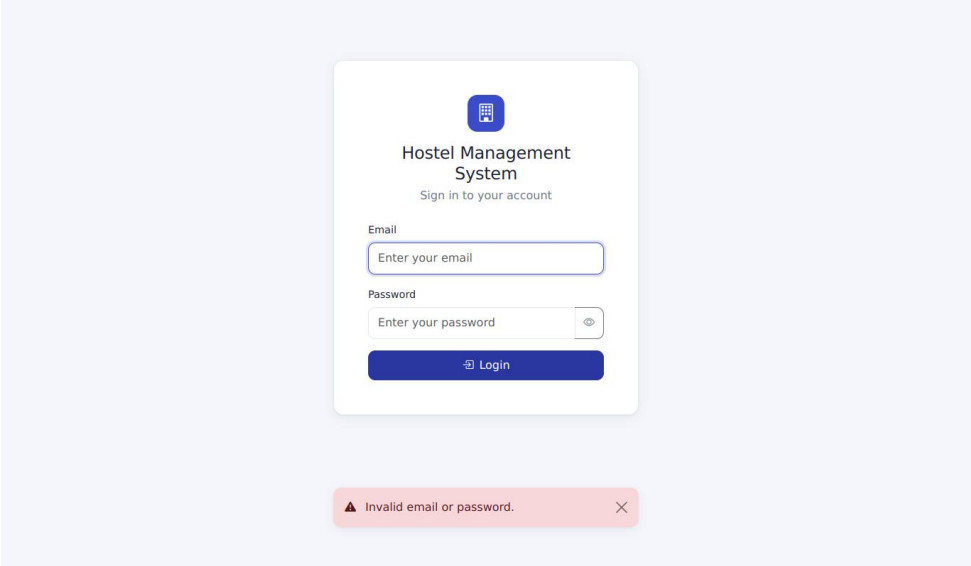
Preconditions: Not logged in.

Steps: 1. Open the login page
2. Enter nobody@example.com with any password
3. Click Login

Input Data: nobody@example.com / x

Expected Result: Same error "Invalid email or password." (the message must not reveal whether the account exists).

AUTH-05_1.jpg — Unknown user: the same generic error message
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

6. AUTH-06 — Logout ends the session

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P1 / UI only

Module: Authentication

Requirement: RA-11

Technique: UI only

Priority: P1

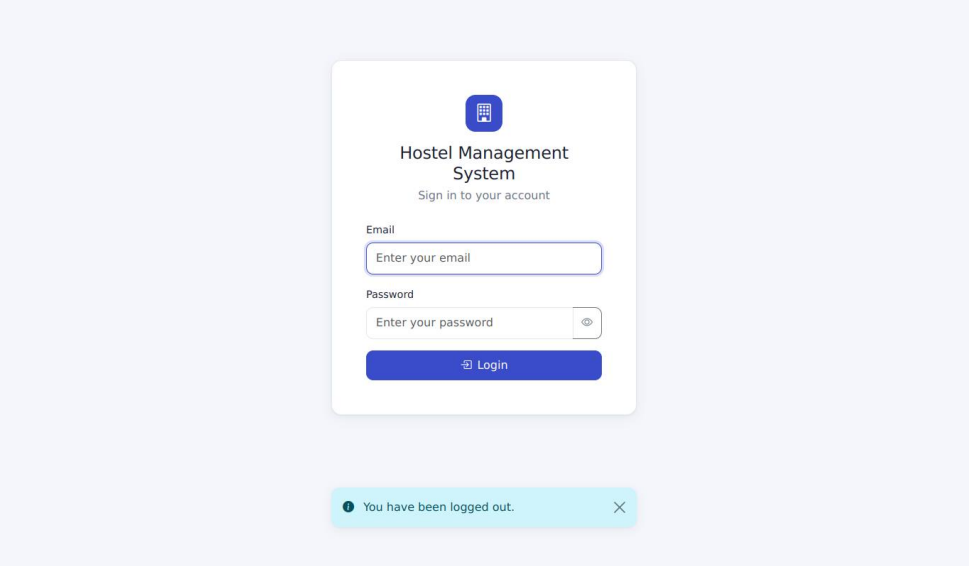
Scenario: Logout ends the session

Preconditions: Logged in as admin.

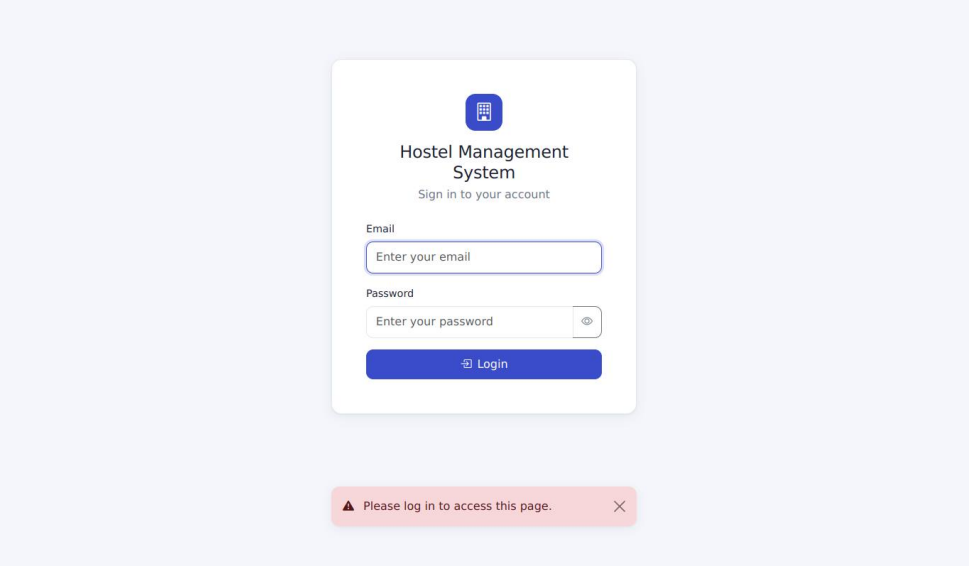
Steps: 1. Click Logout at the bottom of the sidebar
2. In the address bar open http://127.0.0.1:5000/allocation/

Expected Result: After step 1: login page with "You have been logged out."; after step 2: redirected to the login page with "Please log in to access this page."

AUTH-06_1.jpg — Step 1: after clicking Logout - login page with "You have been logged out."
Reference screenshot of expected state; not verified execution evidence.



AUTH-06_2.jpg — Step 2: opening /allocation/ while logged out redirects to the login page
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

7. AUTH-07 — Anonymous access is blocked

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P1 / UI only

Module: Authentication

Requirement: RA-11

Technique: UI only

Priority: P1

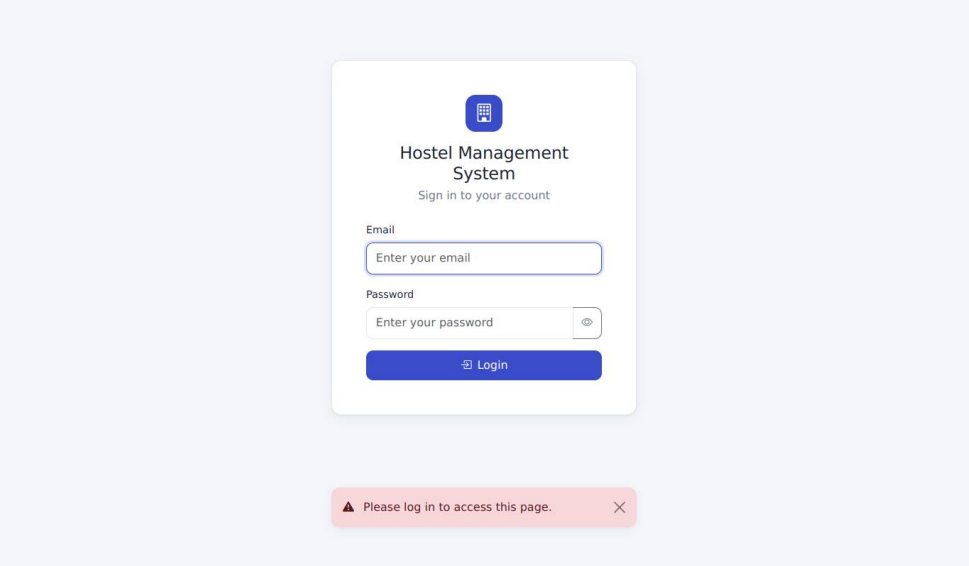
Scenario: Anonymous access is blocked

Preconditions: Not logged in.

Steps: 1. Open http://127.0.0.1:5000/
2. Open /allocation/
3. Open /rooms

Expected Result: Every URL redirects to the login page with "Please log in to access this page."

AUTH-07_1.jpg — Anonymous access to /rooms is redirected to the login page with a message
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

8. AUTH-08 — Login requires a CSRF token

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P1 / UI + browser DevTools

Module: Authentication

Requirement: RA-12

Technique: UI + browser DevTools

Priority: P1

Scenario: Login requires a CSRF token

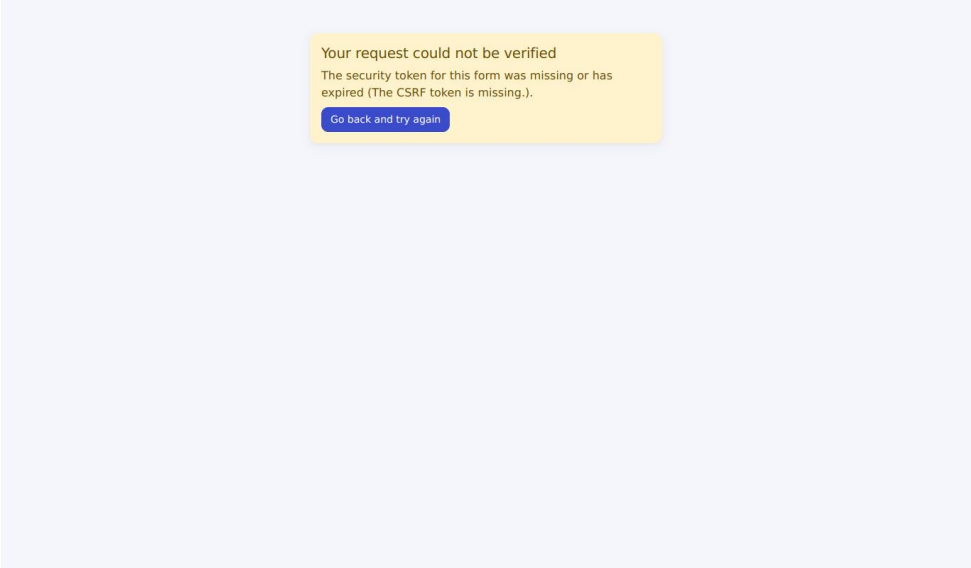
Preconditions: Not logged in; browser DevTools open.

Steps: 1. Open the login page
2. In DevTools > Elements delete the hidden <input name="csrf_token">
3. Enter valid admin credentials and click Login

Input Data: admin@hms.test / admin123

Expected Result: The server rejects the request: page titled "Your request could not be verified" (HTTP 400, visible in the DevTools Network tab). You are not logged in.

AUTH-08_1.jpg — Login without the CSRF token: request rejected (HTTP 400 page)
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

9. AUTH-09 — Lockout after repeated failed logins

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P2 / UI only

Module: Authentication

Requirement: RA-11

Technique: UI only

Priority: P2

Scenario: Lockout after repeated failed logins

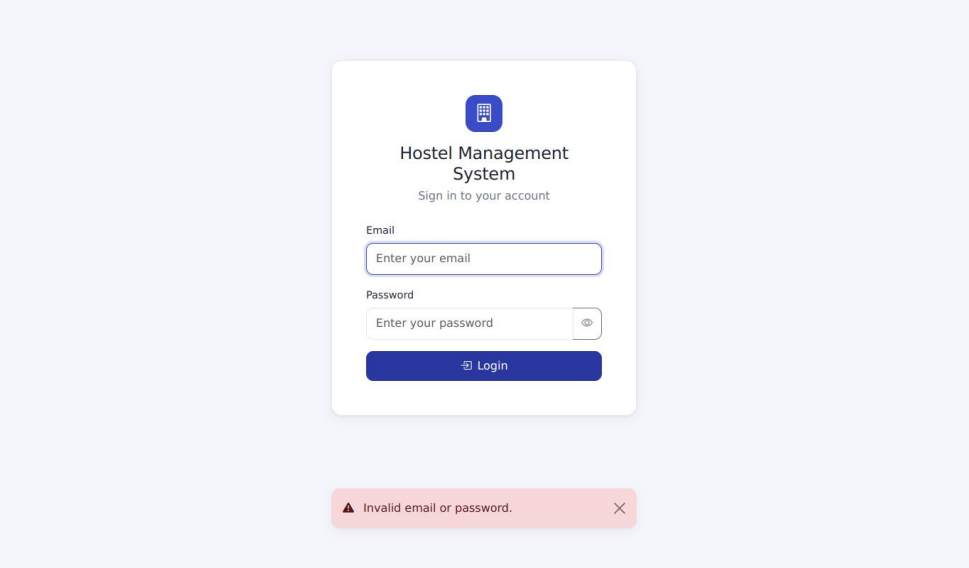
Preconditions: Restart the server with LOGIN_MAX_ATTEMPTS=5 (TESTING_PROCEDURE.md section 3). Not logged in.

Steps: 1. Submit the login form 5 times with dana@student.test and a wrong password
2. Submit a 6th time with the CORRECT password (student123)
3. Log in with john@student.test / student123

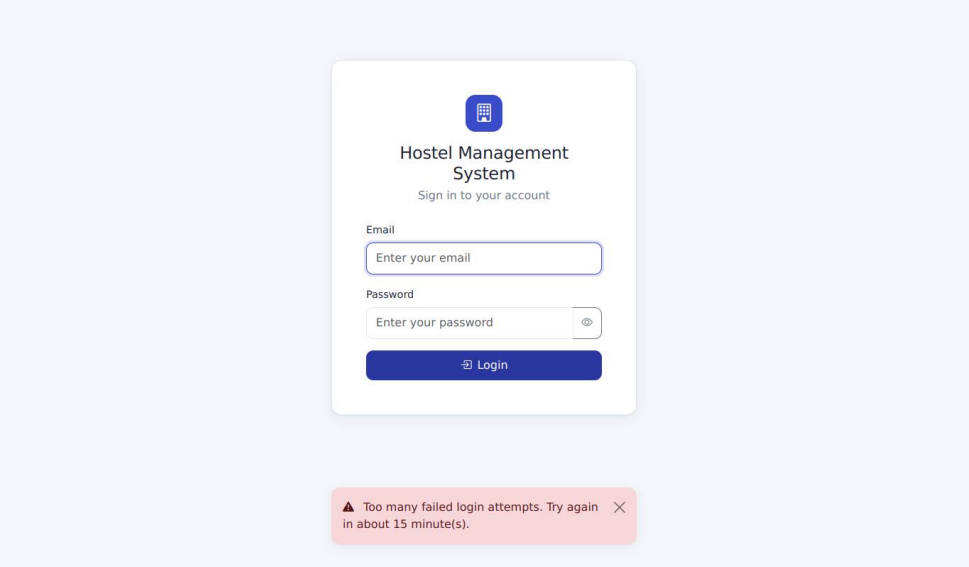
Input Data: dana@student.test

Expected Result: Attempts 1-5: "Invalid email or password."; attempt 6 (even with the correct password): "Too many failed login attempts. Try again in about 15 minute(s)."; step 3 (a different account) still works.

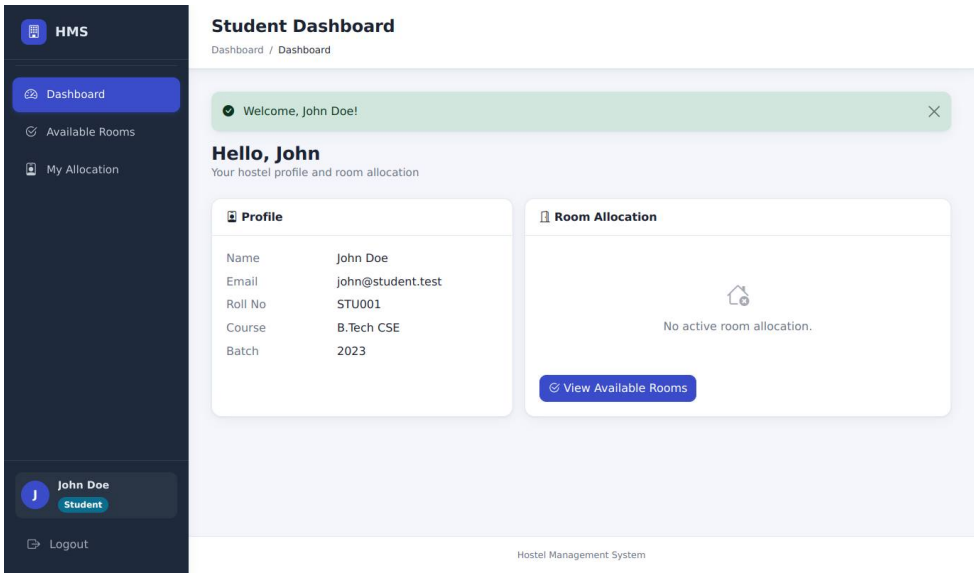
AUTH-09_1.jpg — Attempt 5 of 5 (wrong password): still the normal error message
Reference screenshot of expected state; not verified execution evidence.



AUTH-09_2.jpg — Attempt 6 with the CORRECT password: locked out message
Reference screenshot of expected state; not verified execution evidence.



AUTH-09_3.jpg — A different account (John) can still log in
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

10. AUTH-10 — Logout is not possible via a link (GET)

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P1 / UI only

Module: Authentication

Requirement: RA-11, RA-12

Technique: UI only

Priority: P1

Scenario: Logout is not possible via a link (GET)

Preconditions: Logged in as admin.

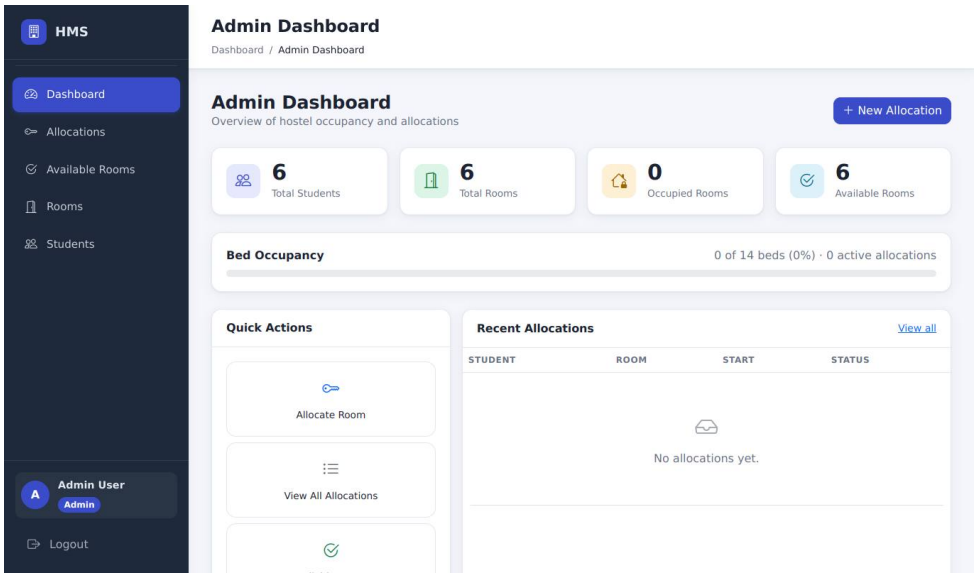
Steps: 1. In the address bar open http://127.0.0.1:5000/auth/logout
2. Open http://127.0.0.1:5000/

Expected Result: Step 1 shows "Method Not Allowed" (HTTP 405). Step 2 shows the dashboard: you are still logged in.

AUTH-10_1.jpg — Step 1: opening /auth/logout in the address bar - "Method Not Allowed"
Reference screenshot of expected state; not verified execution evidence.



AUTH-10_2.jpg — Step 2: the dashboard is still shown - you are still logged in
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

11. ALLOC-01 — Allocate a student, open-ended

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P1 / UI only

Module: Allocation

Requirement: RA-01

Technique: UI only

Priority: P1

Scenario: Allocate a student, open-ended

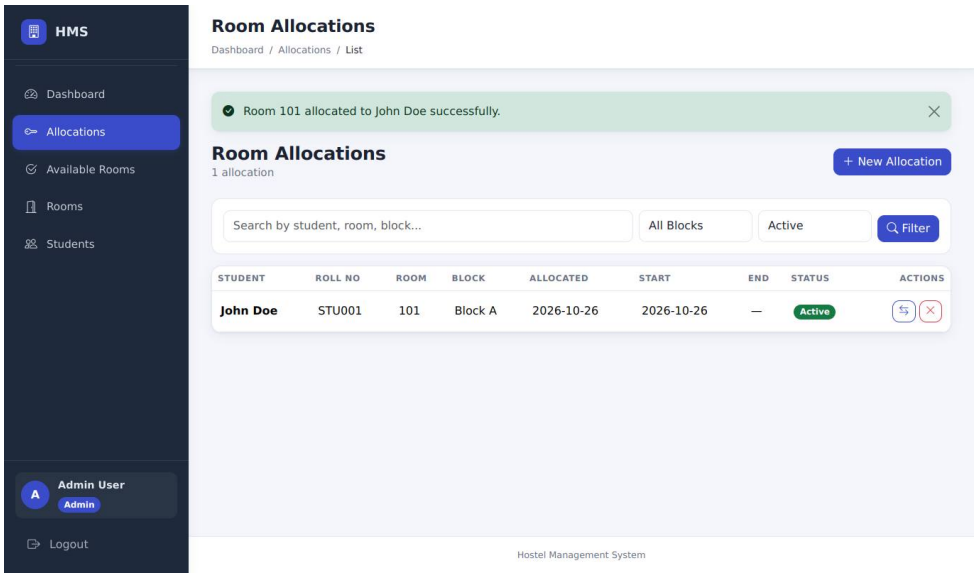
Preconditions: Logged in as admin. Fresh demo data without allocations. No allocations exist.

Steps: 1. Sidebar > Allocations > New Allocation
2. Student: John Doe; Room: Block A, room 101
3. Leave the start date at today; leave End date empty
4. Click Create allocation

Input Data: John Doe -> Block A / 101

Expected Result: Redirected to the allocation list; message "Room 101 allocated to John Doe successfully."; list shows John Doe, Block A, room 101, Status Active, End "—".

ALLOC-01_1.jpg — John Doe allocated to Block A 101 (open-ended): success message and the new row
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

12. ALLOC-08 — Allocate with an end date

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P2 / UI only

Module: Allocation

Requirement: RA-01

Technique: UI only

Priority: P2

Scenario: Allocate with an end date

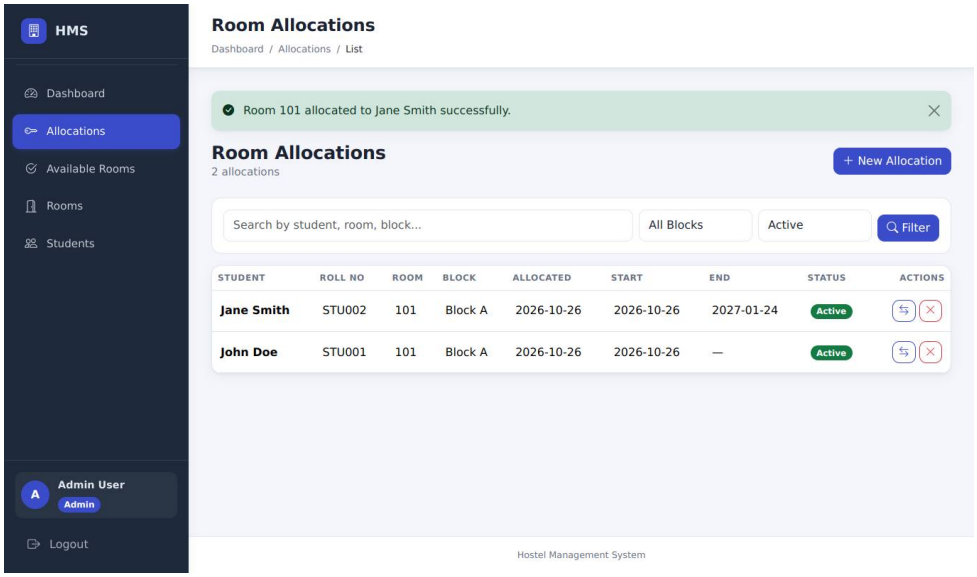
Preconditions: ALLOC-01 done.

Steps: 1. New Allocation
2. Student: Jane Smith; Room: Block A, room 101
3. Start: today; End: today + 90 days
4. Create allocation

Input Data: Jane Smith -> Block A / 101

Expected Result: Success message for room 101 and Jane Smith; list shows the end date. Room Block A 101 is now 2/2.

ALLOC-08_1.jpg — Jane Smith allocated with an end date (90 days ahead)
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

13. OCC-01 — Room becomes Full at capacity

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P1 / UI only

Module: Occupancy

Requirement: RA-04

Technique: UI only

Priority: P1

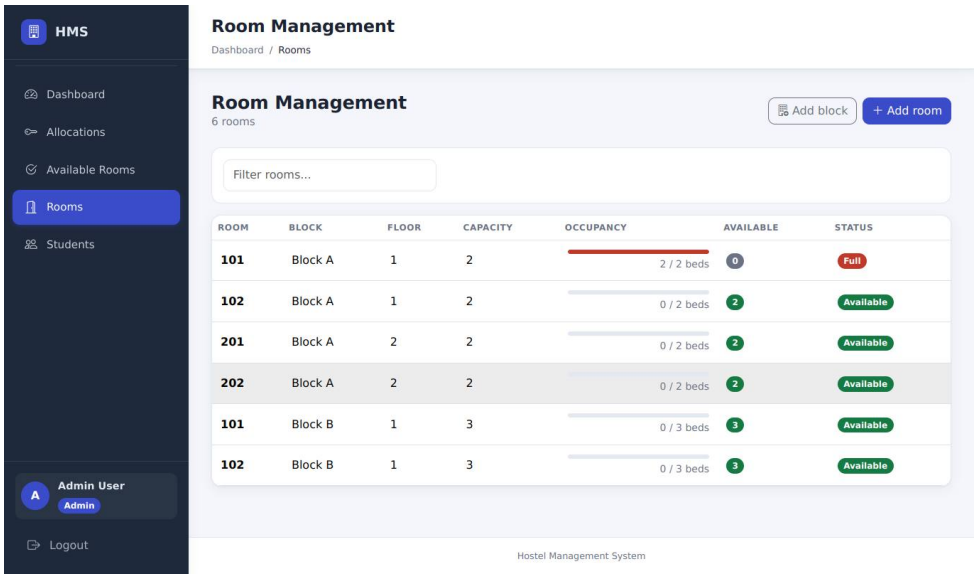
Scenario: Room becomes Full at capacity

Preconditions: ALLOC-01 and ALLOC-08 done.

Steps: 1. Sidebar > Rooms
2. Find Block A room 101

Expected Result: Block A 101 shows 2 of 2 beds occupied and status Full.

OCC-01_1.jpg — Rooms page: Block A 101 is 2 / 2 beds and Full
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

14. ALLOC-05 — Student with an active allocation cannot get a second

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P1 / UI only

Module: Allocation

Requirement: RA-03

Technique: UI only

Priority: P1

Scenario: Student with an active allocation cannot get a second

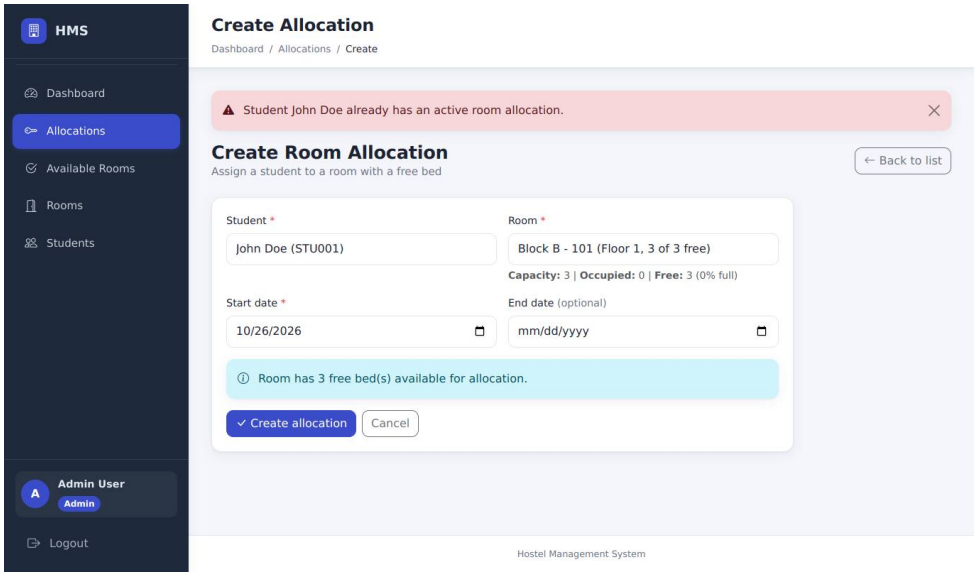
Preconditions: ALLOC-01 done (John has an active allocation).

Steps: 1. New Allocation
2. Student: John Doe; Room: Block B, room 101
3. Create allocation

Input Data: John Doe -> Block B / 101

Expected Result: Error "Student John Doe already has an active room allocation."; no new row in the allocation list.

ALLOC-05_1.jpg — Second allocation for John refused: "already has an active room allocation"
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

15. ALLOC-15 — Form keeps your input after a server-side error

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P2 / UI only

Module: Allocation

Requirement: RA-02, UI-03

Technique: UI only

Priority: P2

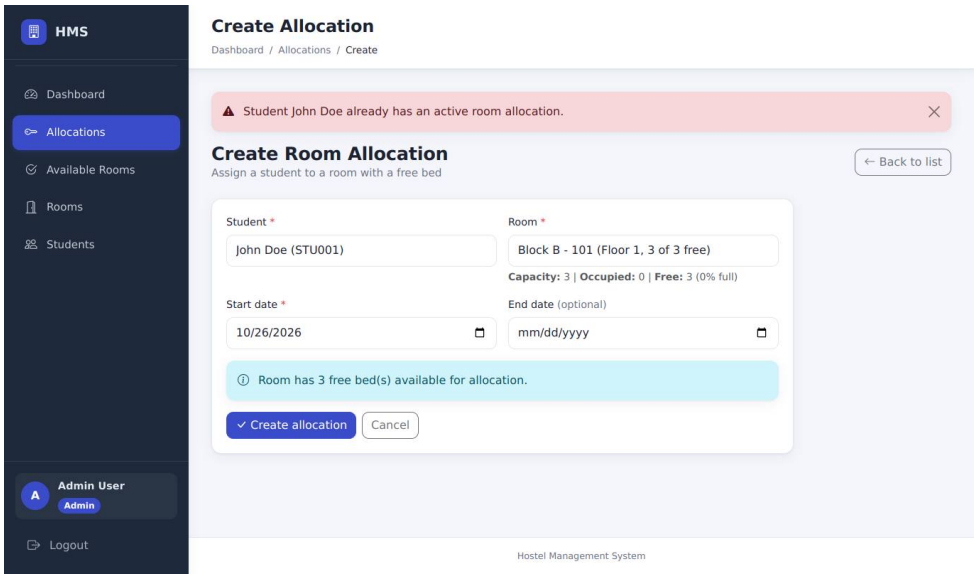
Scenario: Form keeps your input after a server-side error

Preconditions: Immediately after ALLOC-05.

Steps: 1. Look at the New Allocation form that is displayed together with the error

Expected Result: The form is shown again with Student = John Doe, Room = Block B 101 and the start date still selected (you do not have to re-enter everything).

ALLOC-15_1.jpg — The same screen: the form is shown again with John Doe and Block B 101 still selected
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

16. ALLOC-11 — Start date equal to end date is accepted

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P2 / UI only

Module: Allocation

Requirement: RA-01

Technique: UI only

Priority: P2

Scenario: Start date equal to end date is accepted

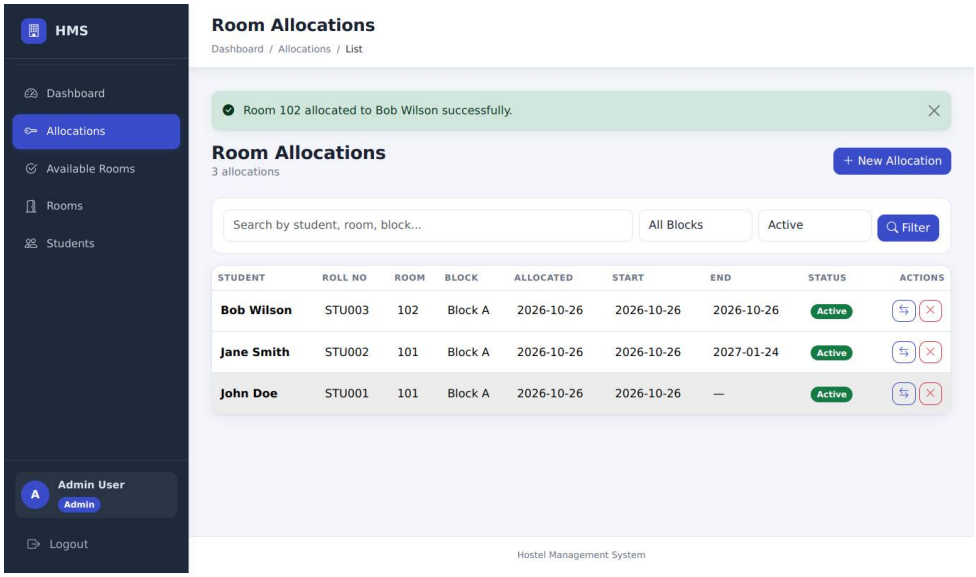
Preconditions: Logged in as admin.

Steps: 1. New Allocation
2. Student: Bob Wilson; Room: Block A, room 102
3. Start: today; End: today
4. Create allocation

Input Data: Bob Wilson -> Block A / 102

Expected Result: Success message "Room 102 allocated to Bob Wilson successfully."

ALLOC-11_1.jpg — Bob Wilson allocated with start date = end date = today
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

17. ALLOC-02 — Allocating to a room that filled up meanwhile is refused

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P1 / UI + browser DevTools

Module: Allocation

Requirement: RA-04

Technique: UI + browser DevTools

Priority: P1

Scenario: Allocating to a room that filled up meanwhile is refused

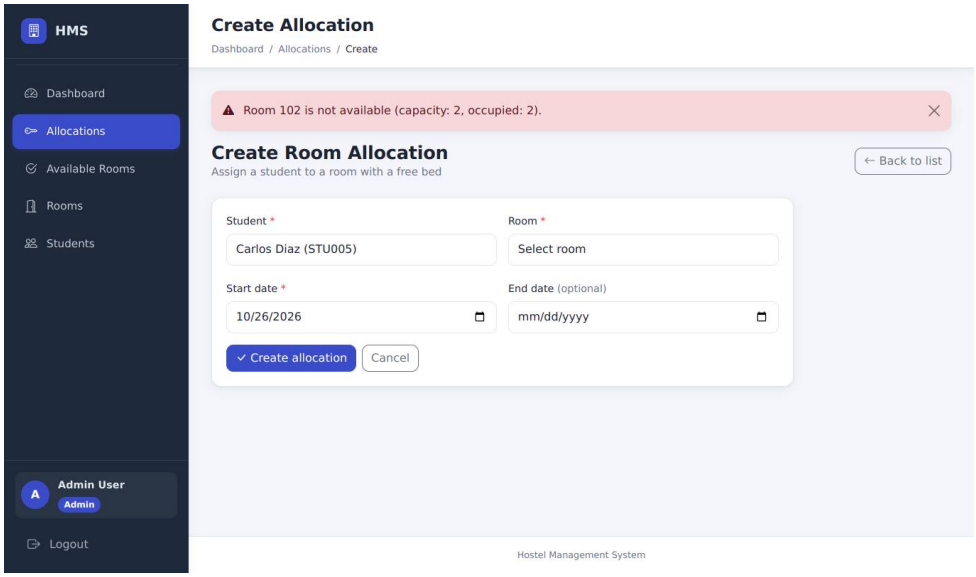
Preconditions: Block A 102 has exactly 1 free bed (Bob only). Two browser tabs on the New Allocation form, both opened BEFORE step 1 (both list Block A 102).

Steps: 1. Tab 1: allocate Alice Brown to Block A 102 and submit (succeeds, room now full)
2. Tab 2 (stale form, still lists room 102): select Carlos Diaz and Block A 102, submit

Input Data: Carlos Diaz -> Block A / 102 (stale form)

Expected Result: Tab 2 shows error "Room 102 is not available (capacity: 2, occupied: 2)."; Carlos has no allocation; room 102 stays 2/2.

ALLOC-02_1.jpg — Tab 2 (stale form): room 102 is no longer available - the server refuses the allocation
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

18. ALLOC-09 — Warden can create an allocation

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P2 / UI only

Module: Allocation

Requirement: RA-01

Technique: UI only

Priority: P2

Scenario: Warden can create an allocation

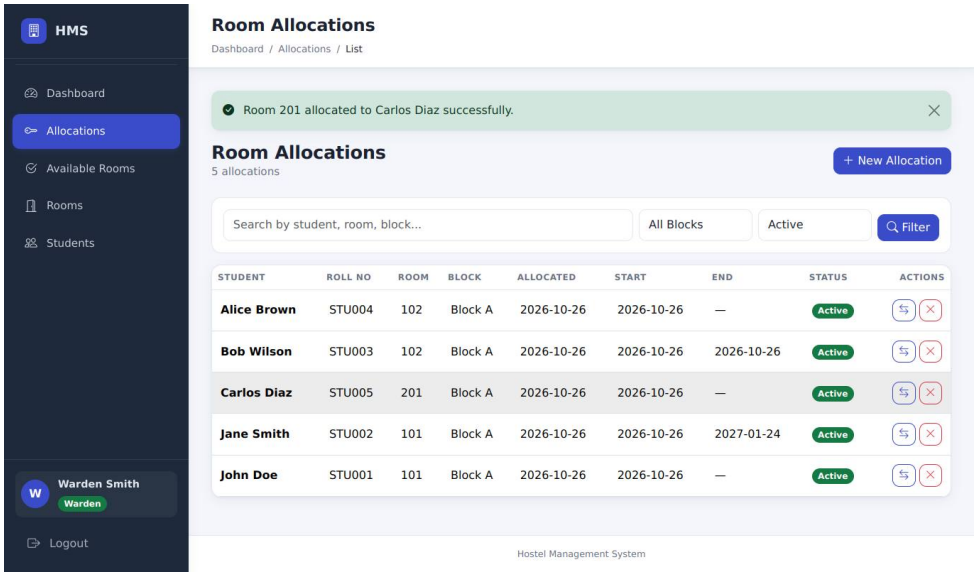
Preconditions: Logged in as WARDEN (log out first).

Steps: 1. New Allocation
2. Student: Carlos Diaz; Room: Block A, room 201
3. Create allocation

Input Data: warden@hms.test / warden123 / Carlos Diaz -> Block A / 201

Expected Result: Success message "Room 201 allocated to Carlos Diaz successfully."

ALLOC-09_1.jpg — Warden allocates Carlos Diaz to Block A 201
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

19. ALLOC-03 — Non-existent student id is rejected

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P2 / UI + browser DevTools

Module: Allocation

Requirement: RA-02

Technique: UI + browser DevTools

Priority: P2

Scenario: Non-existent student id is rejected

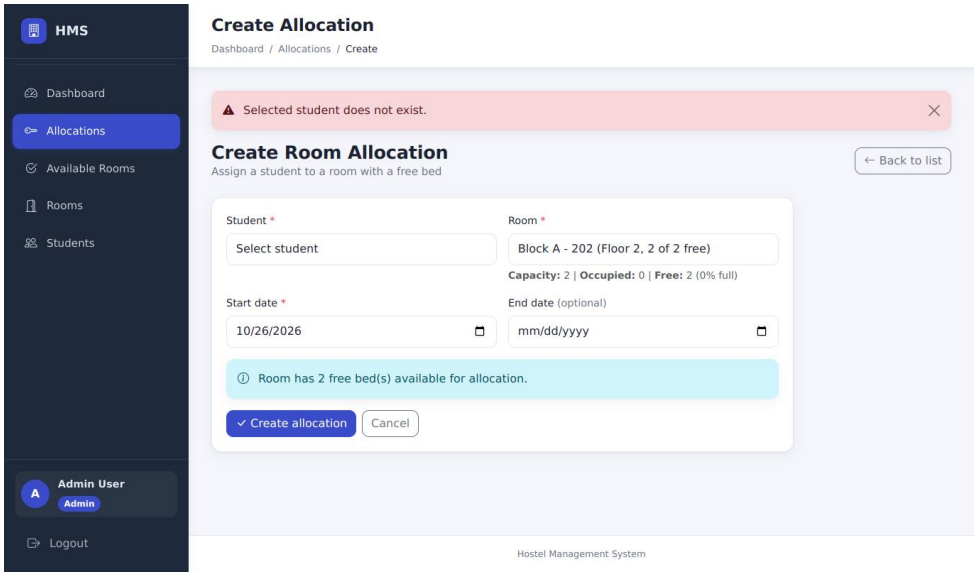
Preconditions: Logged in as admin.

Steps: 1. Open New Allocation
2. In DevTools change the value of one <option> in the Student select to 99999
3. Select that option, choose a room, submit

Input Data: student_id=99999

Expected Result: Error "Selected student does not exist."; nothing created.

ALLOC-03_1.jpg — Student id 99999 (edited in DevTools): "Selected student does not exist."
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

20. ALLOC-04 — Non-existent room id is rejected

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P2 / UI + browser DevTools

Module: Allocation

Requirement: RA-02

Technique: UI + browser DevTools

Priority: P2

Scenario: Non-existent room id is rejected

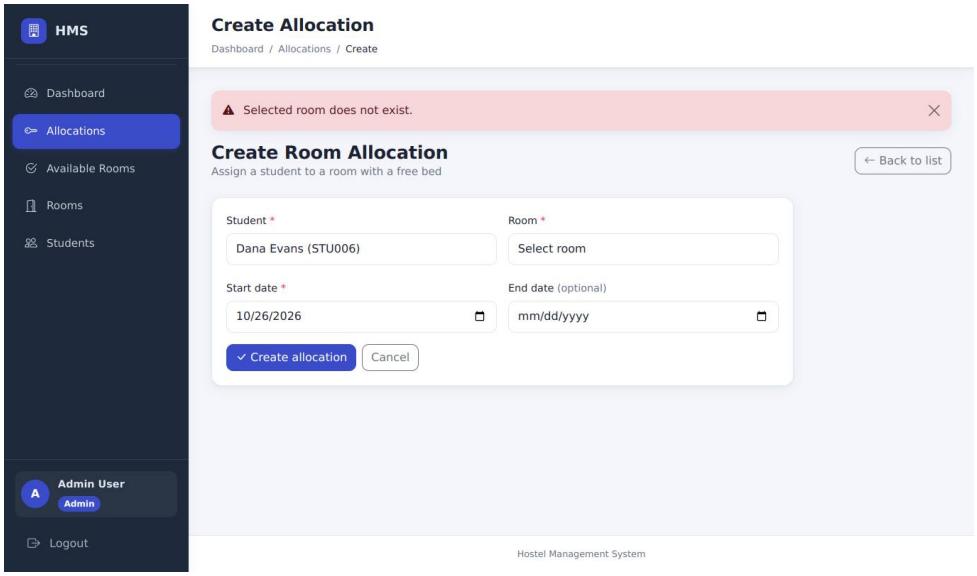
Preconditions: Logged in as admin.

Steps: 1. Open New Allocation
2. In DevTools change the value of one <option> in the Room select to 99999
3. Select a student and that room option, submit

Input Data: room_id=99999

Expected Result: Error "Selected room does not exist."; nothing created.

ALLOC-04_1.jpg — Room id 99999 (edited in DevTools): "Selected room does not exist."
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

21. ALLOC-06 — Required fields

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P1 / UI only

Module: Allocation

Requirement: RA-02

Technique: UI only

Priority: P1

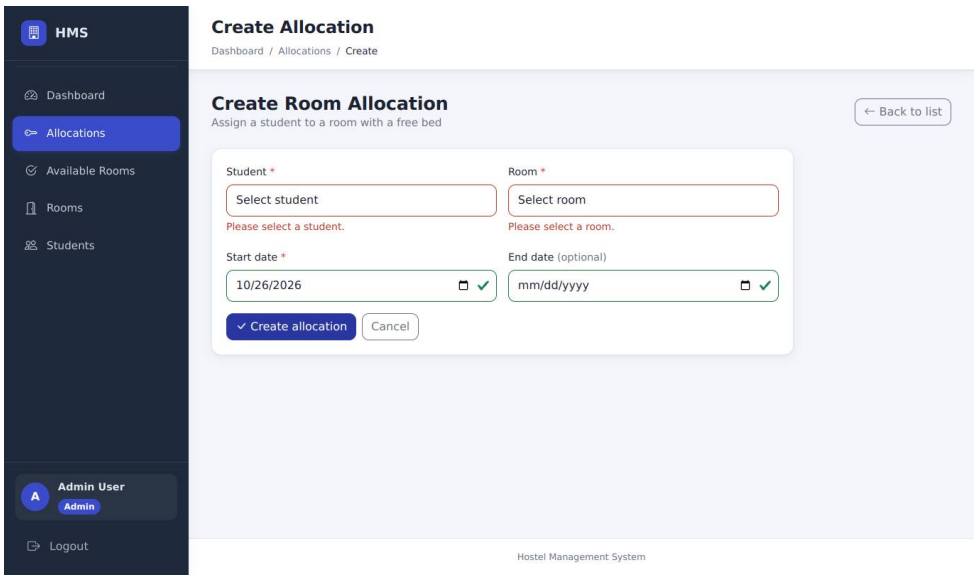
Scenario: Required fields

Preconditions: Logged in as admin.

Steps: 1. Open New Allocation
2. Leave Student and Room unselected
3. Click Create allocation

Expected Result: The form is not submitted: the empty required fields are highlighted/flagged and the Create allocation button is still clickable (it must not stay stuck on "Processing..."). (If you bypass the browser check the server answers "Student, room, and start date are required.")

ALLOC-06_1.jpg — Submitting with Student and Room empty: the required fields are flagged and nothing is sent
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

22. ALLOC-07 — Invalid date text is rejected

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P2 / UI + browser DevTools

Module: Allocation

Requirement: RA-02

Technique: UI + browser DevTools

Priority: P2

Scenario: Invalid date text is rejected

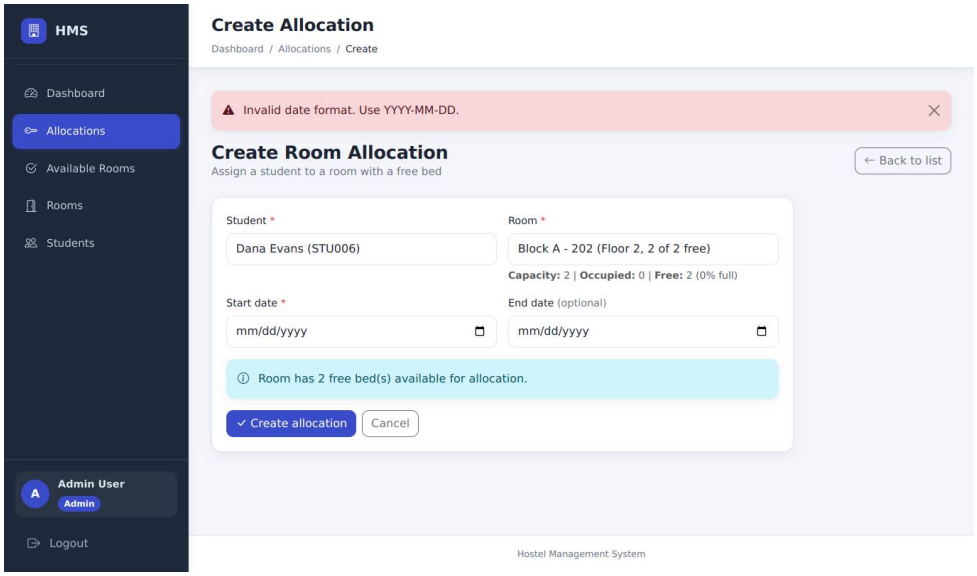
Preconditions: Logged in as admin.

Steps: 1. Open New Allocation
2. In DevTools change the Start date <input type="date"> to type="text"
3. Type 2026-13-45; select student and room; submit

Input Data: 2026-13-45

Expected Result: Error "Invalid date format. Use YYYY-MM-DD."; nothing created; entered values kept.

ALLOC-07_1.jpg — Start date "2026-13-45" (field changed to text in DevTools): "Invalid date format"
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

23. ALLOC-10 — End date before start date is rejected

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P1 / UI only

Module: Allocation

Requirement: RA-02

Technique: UI only

Priority: P1

Scenario: End date before start date is rejected

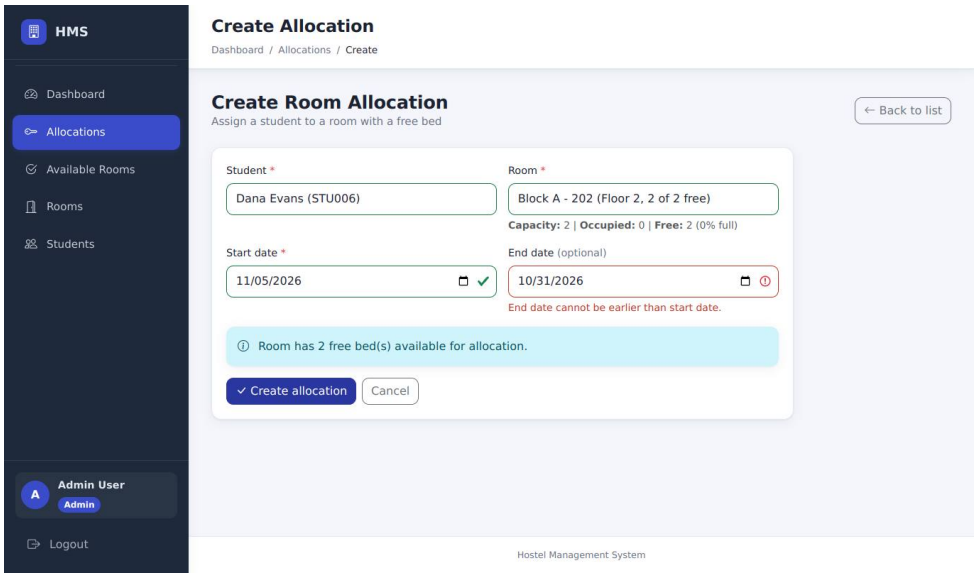
Preconditions: Logged in as admin.

Steps: 1. Open New Allocation
2. Choose a student without allocation (Dana Evans) and any available room
3. Start: today + 10 days; End: today + 5 days
4. Submit

Input Data: start > end

Expected Result: Submission refused: either the browser/JS shows a validation message ("End date cannot be earlier than start date.") or the server shows it; the button stays clickable; no allocation created.

ALLOC-10_1.jpg — End date (today+5) before start date (today+10): the form refuses to submit
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

24. ALLOC-14 — Dates outside 2000-2100 are rejected

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P2 / UI + browser DevTools

Module: Allocation

Requirement: RA-02

Technique: UI + browser DevTools

Priority: P2

Scenario: Dates outside 2000-2100 are rejected

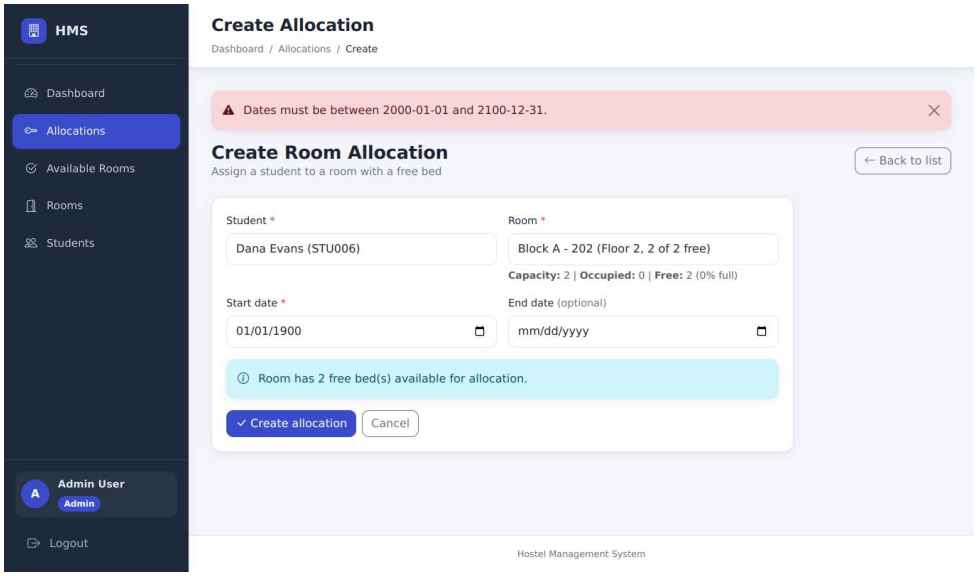
Preconditions: Logged in as admin.

Steps: 1. Open New Allocation
2. In DevTools change Start date to type="text" and enter 1900-01-01; choose Dana Evans and a room; submit
3. Repeat with 9999-12-31

Input Data: 1900-01-01 / 9999-12-31

Expected Result: Error "Dates must be between 2000-01-01 and 2100-12-31." both times; nothing created.

ALLOC-14_1.jpg — Start date 1900-01-01: "Dates must be between 2000-01-01 and 2100-12-31."
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

25. ALLOC-13 — Create requires a CSRF token

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P1 / UI + browser DevTools

Module: Allocation

Requirement: RA-12

Technique: UI + browser DevTools

Priority: P1

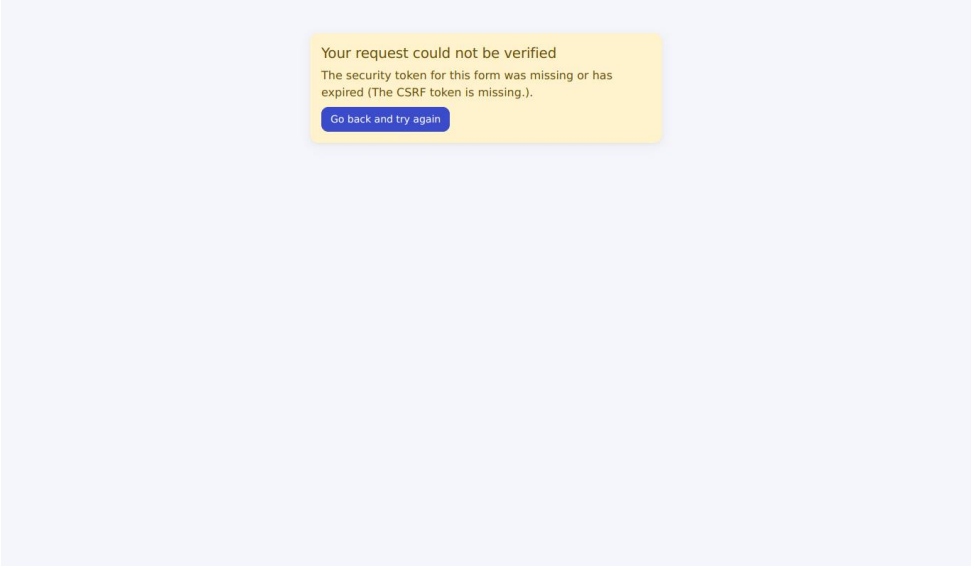
Scenario: Create requires a CSRF token

Preconditions: Logged in as admin.

Steps: 1. Open New Allocation and fill it with valid data (Dana Evans, Block A 202)
2. In DevTools delete the hidden <input name="csrf_token">
3. Submit

Expected Result: Page "Your request could not be verified" (HTTP 400); no allocation created for Dana Evans.

ALLOC-13_1.jpg — CSRF token deleted in DevTools: "Your request could not be verified" (HTTP 400)
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

26. REALLOC-01 — Reallocate to a room with a free bed

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P1 / UI only

Module: Reallocation

Requirement: RA-05

Technique: UI only

Priority: P1

Scenario: Reallocate to a room with a free bed

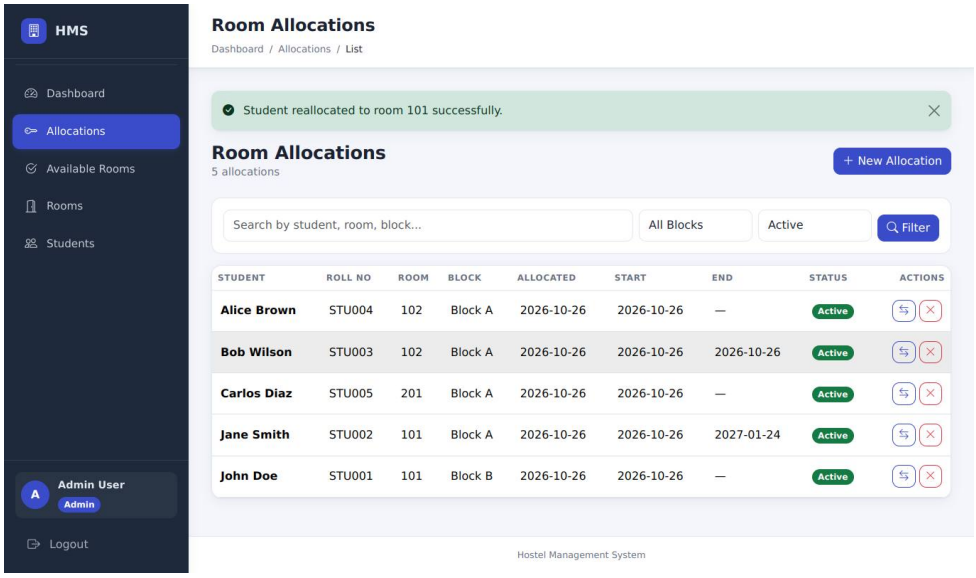
Preconditions: John has an active allocation in Block A 101 (full 2/2).

Steps: 1. Allocations list > click the Reallocate icon on John Doe
2. New room: Block B, room 101
3. Click Reallocate

Input Data: John: A/101 -> B/101

Expected Result: Message "Student reallocated to room 101 successfully."; list shows John in Block B. Rooms page: Block A 101 is now 1/2 and Available; Block B 101 is 1/3.

REALLOC-01_1.jpg — John reallocated to Block B 101: message "Student reallocated to room 101 successfully."
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

27. OCC-02 — Room becomes Available again when a bed is freed

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P1 / UI only

Module: Occupancy

Requirement: RA-04

Technique: UI only

Priority: P1

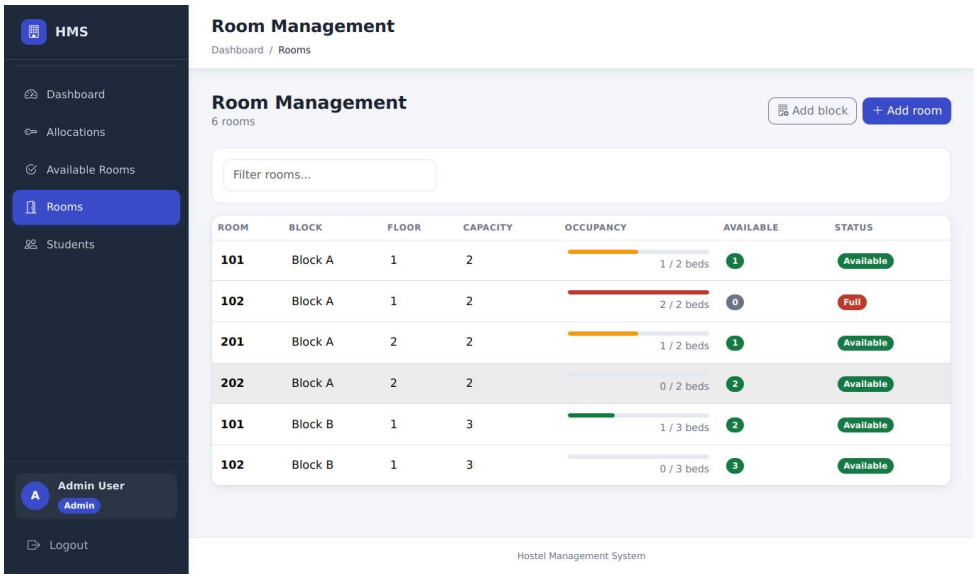
Scenario: Room becomes Available again when a bed is freed

Preconditions: Right after REALLOC-01.

Steps: 1. Sidebar > Rooms
2. Find Block A room 101

Expected Result: Block A 101 shows 1 of 2 beds occupied and status Available.

OCC-02_1.jpg — Rooms page: Block A 101 is now 1 / 2 beds and Available; Block B 101 is 1 / 3
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

28. REALLOC-03 — Full rooms are not offered

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P2 / UI only

Module: Reallocation

Requirement: RA-05, RA-04

Technique: UI only

Priority: P2

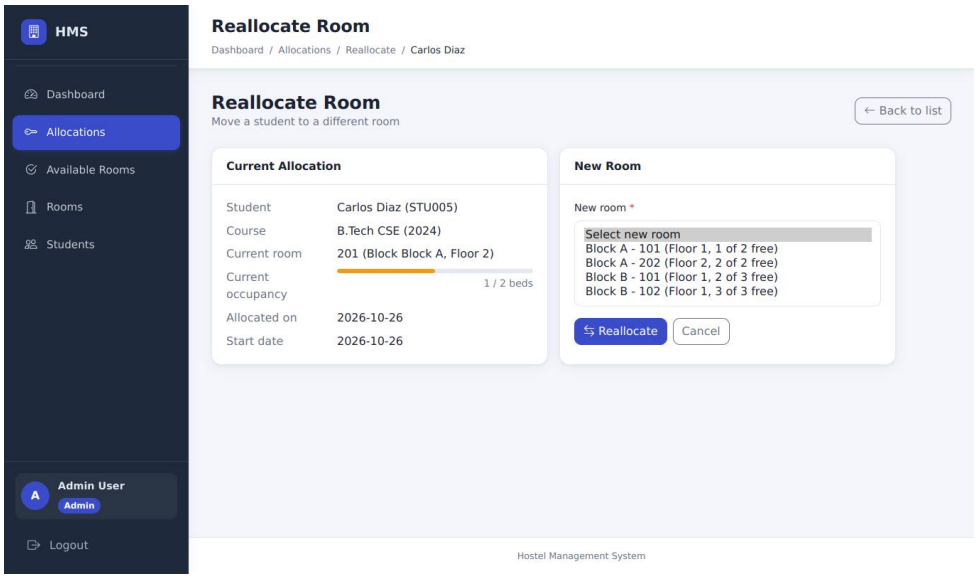
Scenario: Full rooms are not offered

Preconditions: Block A 102 is full (Bob + Alice from ALLOC-02).

Steps: 1. Click Reallocate on Carlos Diaz
2. Open the "New room" drop-down

Expected Result: Block A 102 (full) is not in the list; only rooms with a free bed are offered.

REALLOC-03_1.jpg — Reallocate form for Carlos (list expanded for the screenshot): Block A 102 (full) is not offered
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

29. REALLOC-06 — Reallocating to the same room is refused

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P3 / UI + browser DevTools

Module: Reallocation

Requirement: RA-05

Technique: UI + browser DevTools

Priority: P3

Scenario: Reallocating to the same room is refused

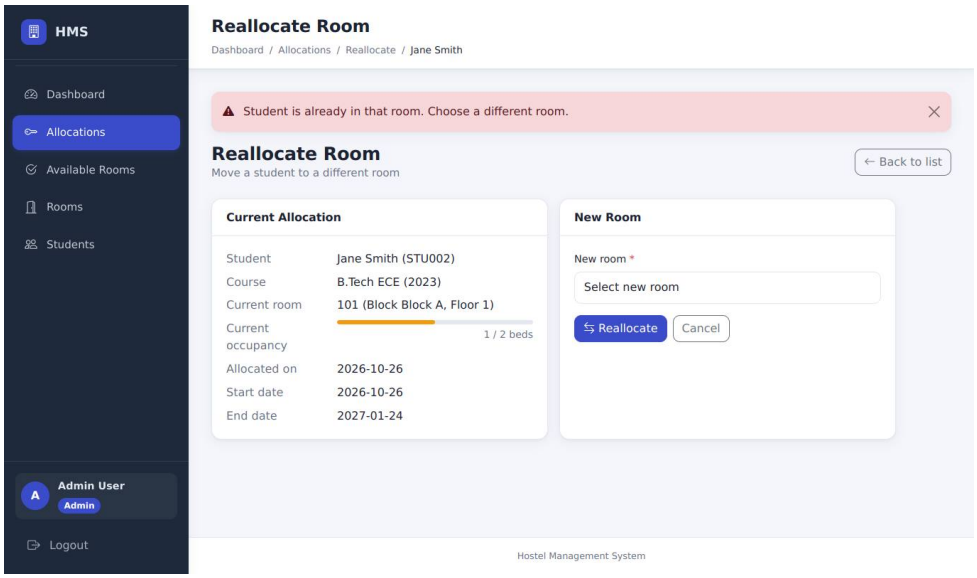
Preconditions: Jane Smith is allocated to Block A 101 (room id 1 in the demo data).

Steps: 1. Click Reallocate on Jane Smith (her current room is not offered in the list)
2. In DevTools > Elements change the value of any one <option> in the "New room" select to 1 (the id of Block A 101)
3. Select that option and click Reallocate

Input Data: new_room_id=1 (her current room)

Expected Result: Error "Student is already in that room. Choose a different room."; no change to the allocation or room counters.

REALLOC-06_1.jpg — Room id 1 = Jane's current room (option value edited in DevTools): "Student is already in that room. Choose a different room."
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

30. REALLOC-04 — Room is required

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P2 / UI only

Module: Reallocation

Requirement: RA-05

Technique: UI only

Priority: P2

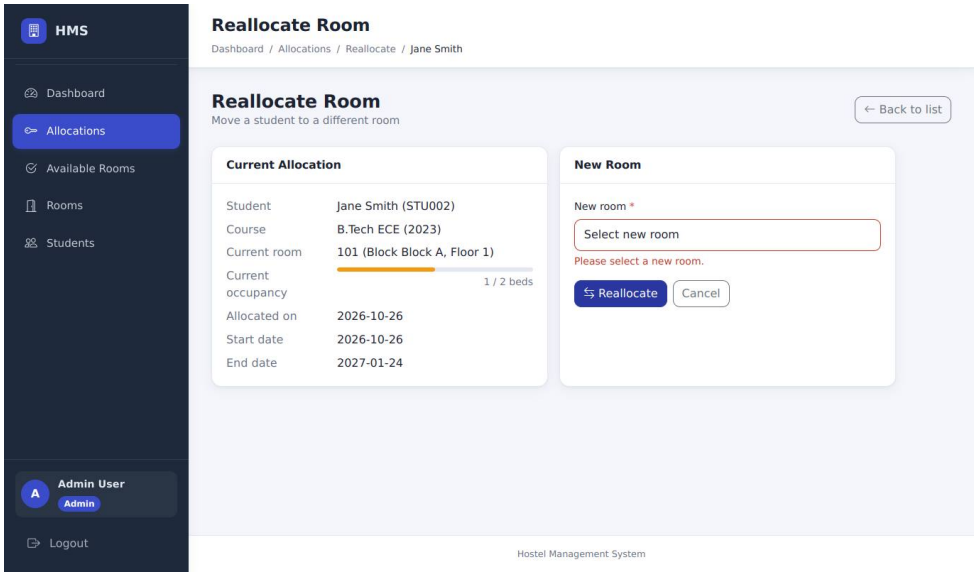
Scenario: Room is required

Preconditions: Reallocate form open.

Steps: 1. Leave "Select new room" selected
2. Click Reallocate

Expected Result: Not submitted (field flagged) and the Reallocate button stays clickable; if bypassed the server says "New room is required."

REALLOC-04_1.jpg — No room selected: the field is flagged and the form is not submitted
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

31. REALLOC-02 — Non-existent room id is rejected

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P3 / UI + browser DevTools

Module: Reallocation

Requirement: RA-05

Technique: UI + browser DevTools

Priority: P3

Scenario: Non-existent room id is rejected

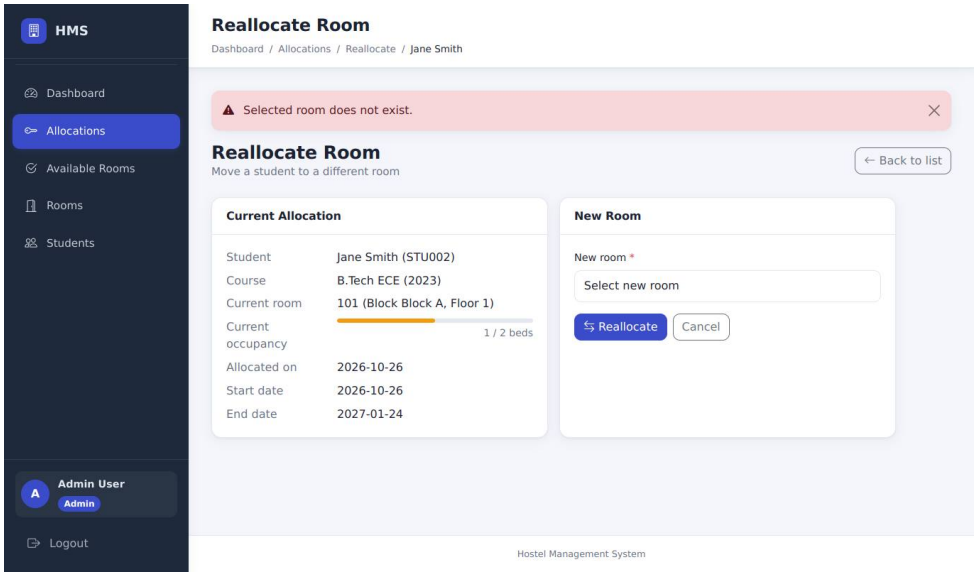
Preconditions: Reallocate form open.

Steps: 1. In DevTools change an <option> value to 99999
2. Select it and click Reallocate

Input Data: room_id=99999

Expected Result: Error "Selected room does not exist."; allocation unchanged.

REALLOC-02_1.jpg — Room id 99999 (edited in DevTools): "Selected room does not exist."
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

32. REALLOC-05 — Reallocate requires a CSRF token

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P1 / UI + browser DevTools

Module: Reallocation

Requirement: RA-12

Technique: UI + browser DevTools

Priority: P1

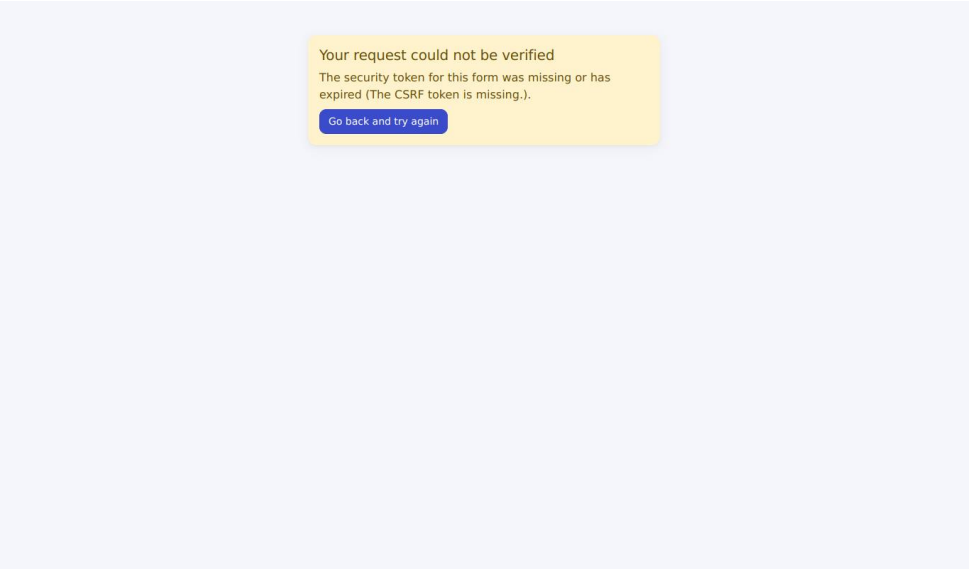
Scenario: Reallocate requires a CSRF token

Preconditions: Reallocate form open.

Steps: 1. Delete the hidden csrf_token input in DevTools
2. Choose a room; click Reallocate

Expected Result: Page "Your request could not be verified" (HTTP 400); allocation unchanged.

REALLOC-05_1.jpg — CSRF token deleted: "Your request could not be verified" (HTTP 400); Jane stays where she is
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

33. CANCEL-02 — "Go back" keeps the allocation

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P1 / UI only

Module: Cancellation

Requirement: RA-06, UI-03

Technique: UI only

Priority: P1

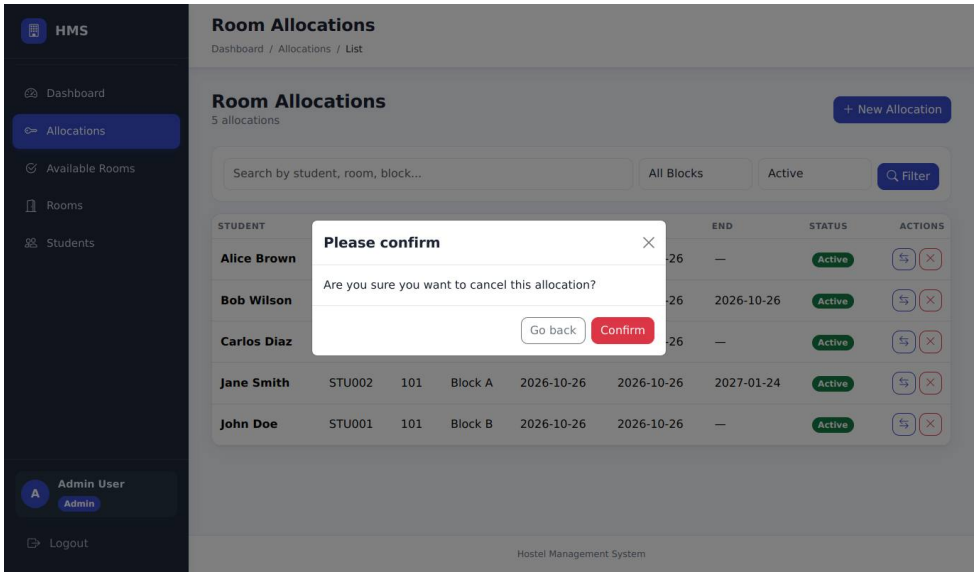
Scenario: "Go back" keeps the allocation

Preconditions: Active allocation exists (Carlos Diaz, Block A 201).

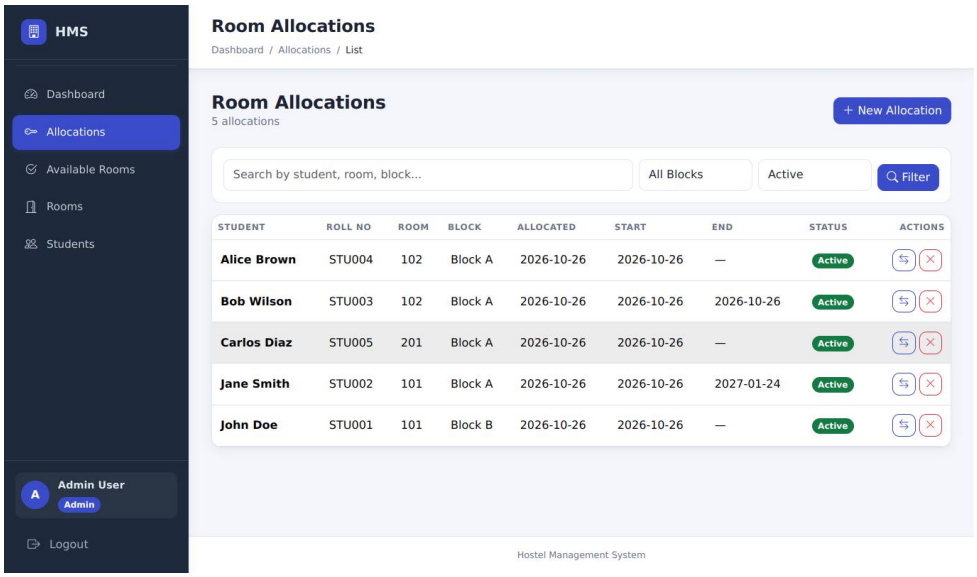
Steps: 1. Allocations list > click the red Cancel (x) icon on Carlos Diaz
2. In the dialog click "Go back"

Expected Result: A modal "Please confirm - Are you sure you want to cancel this allocation?" appears; after Go back the dialog closes and the allocation is still Active.

CANCEL-02_1.jpg — Confirmation dialog for cancelling Carlos Diaz's allocation
Reference screenshot of expected state; not verified execution evidence.



CANCEL-02_2.jpg — After "Go back": the dialog is closed and Carlos is still Active
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

34. CANCEL-01 — Confirm cancels and frees the bed

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P1 / UI only

Module: Cancellation

Requirement: RA-06

Technique: UI only

Priority: P1

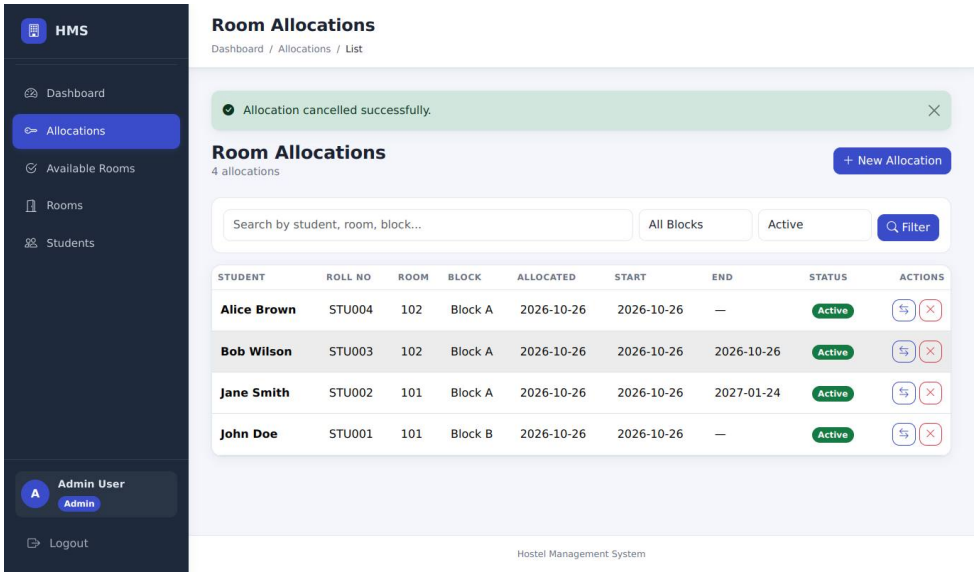
Scenario: Confirm cancels and frees the bed

Preconditions: As CANCEL-02.

Steps: 1. Click the Cancel (x) icon on Carlos Diaz again
2. Click "Confirm"

Expected Result: Message "Allocation cancelled successfully."; Carlos no longer appears under Status = Active; Block A 201 shows 0 of 2 beds.

CANCEL-01_1.jpg — After "Confirm": "Allocation cancelled successfully." and Carlos is gone from the active list
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

35. CANCEL-04 — Cancelled allocations offer no actions

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P2 / UI only

Module: Cancellation

Requirement: RA-06

Technique: UI only

Priority: P2

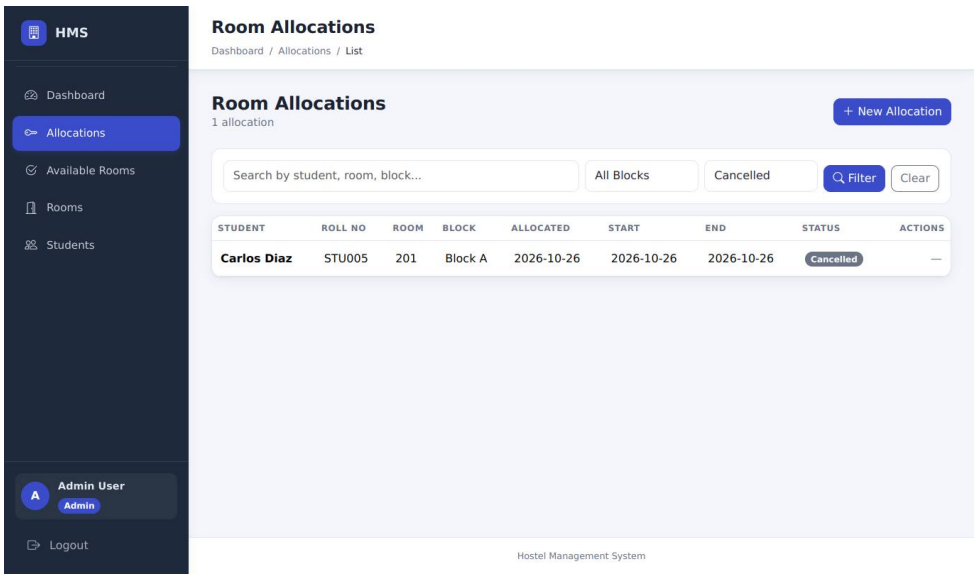
Scenario: Cancelled allocations offer no actions

Preconditions: CANCEL-01 done.

Steps: 1. Allocations list > set Status filter to Cancelled > Filter

Expected Result: Carlos Diaz is listed with status Cancelled and there are no Reallocate/Cancel buttons on that row.

CANCEL-04_1.jpg — Status = Cancelled: Carlos is listed as Cancelled with no action buttons
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

36. CANCEL-03 — Cancel requires a CSRF token

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P1 / UI + browser DevTools

Module: Cancellation

Requirement: RA-12

Technique: UI + browser DevTools

Priority: P1

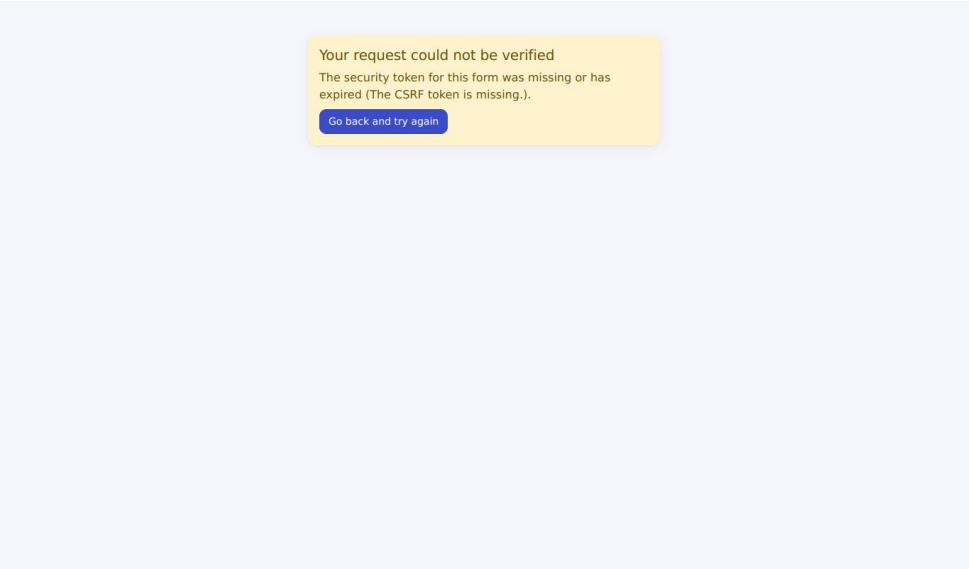
Scenario: Cancel requires a CSRF token

Preconditions: Active allocation exists (e.g. Bob Wilson).

Steps: 1. In DevTools delete the hidden csrf_token input inside the Cancel form of Bob Wilson
2. Click Cancel and Confirm

Expected Result: Page "Your request could not be verified" (HTTP 400); Bob's allocation is still Active.

CANCEL-03_1.jpg — Cancel without a CSRF token: HTTP 400 page; Bob's allocation is untouched
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

37. LIST-01 — Search filters the list

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P2 / UI only

Module: Allocation list

Requirement: RA-07

Technique: UI only

Priority: P2

Scenario: Search filters the list

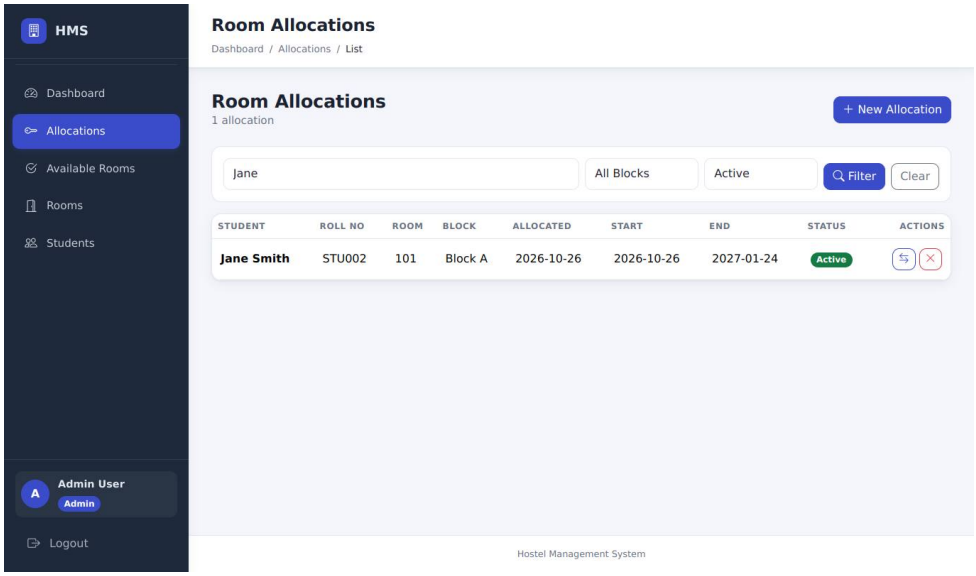
Preconditions: Several active allocations.

Steps: 1. Allocations list > type "Jane" in the search box > Filter
2. Search "nobody-matches"

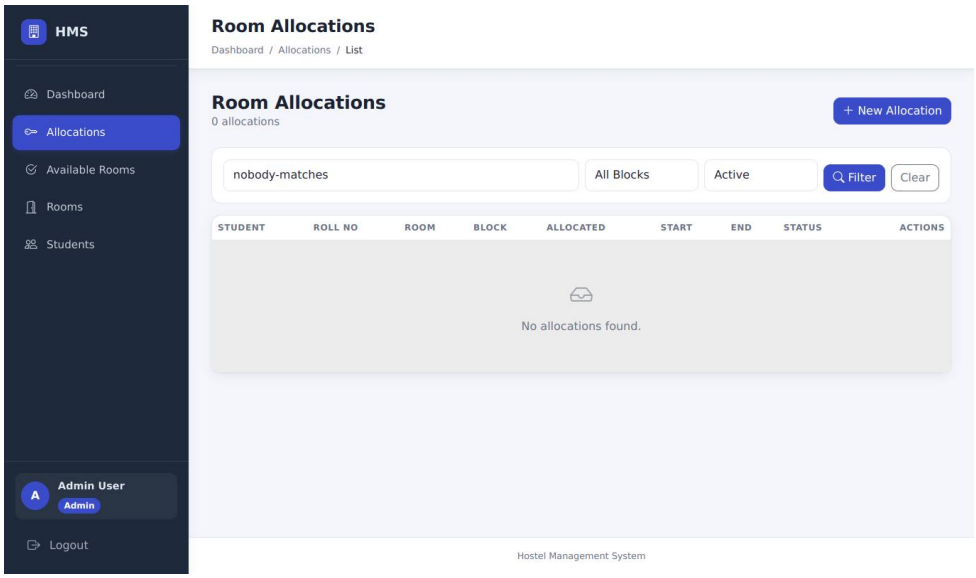
Input Data: Jane / nobody-matches

Expected Result: Step 1: only Jane Smith's row; step 2: "No allocations found."

LIST-01_1.jpg — Search "Jane": only Jane Smith's row
Reference screenshot of expected state; not verified execution evidence.



LIST-01_2.jpg — Search "nobody-matches": "No allocations found."
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

38. LIST-02 — Block and status filters

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P2 / UI only

Module: Allocation list

Requirement: RA-07

Technique: UI only

Priority: P2

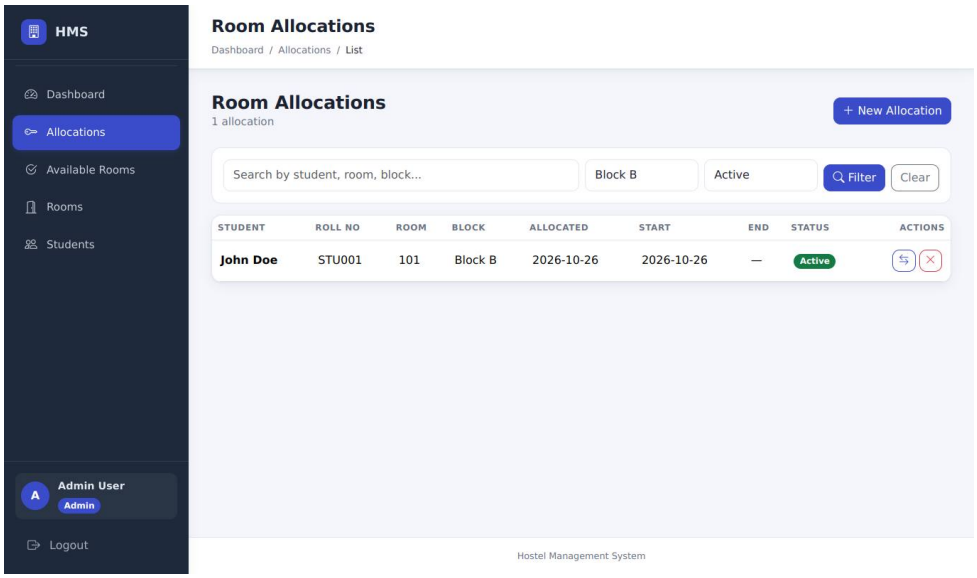
Scenario: Block and status filters

Preconditions: Active and cancelled allocations exist.

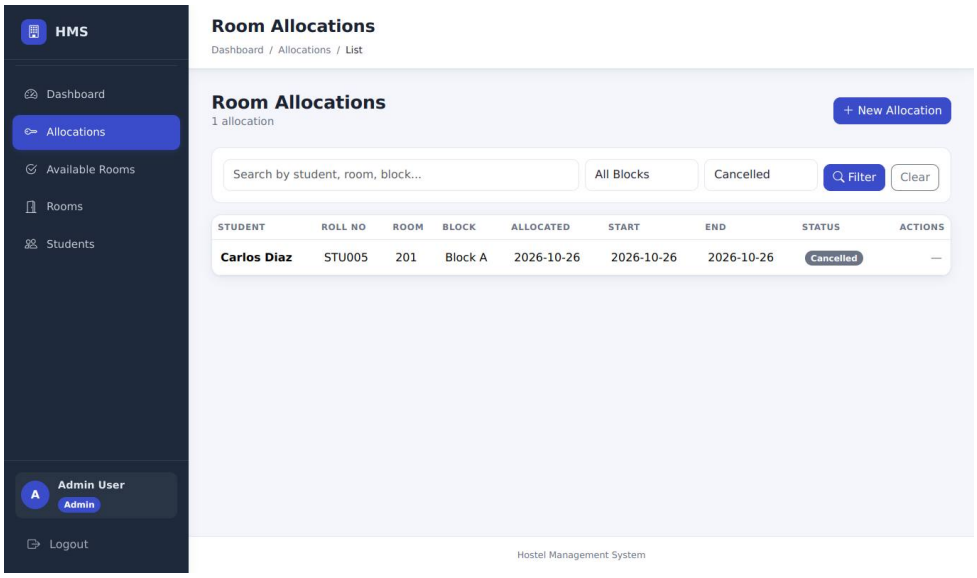
Steps: 1. Select Block B > Filter
2. Select Status Cancelled > Filter

Expected Result: Step 1 lists only Block B rows; step 2 lists only cancelled allocations.

LIST-02_1.jpg — Block B filter: only Block B rows
Reference screenshot of expected state; not verified execution evidence.



LIST-02_2.jpg — Status Cancelled filter: only cancelled allocations
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

39. AVAIL-01 — Only rooms with free beds are listed

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P1 / UI only

Module: Available rooms

Requirement: RA-08

Technique: UI only

Priority: P1

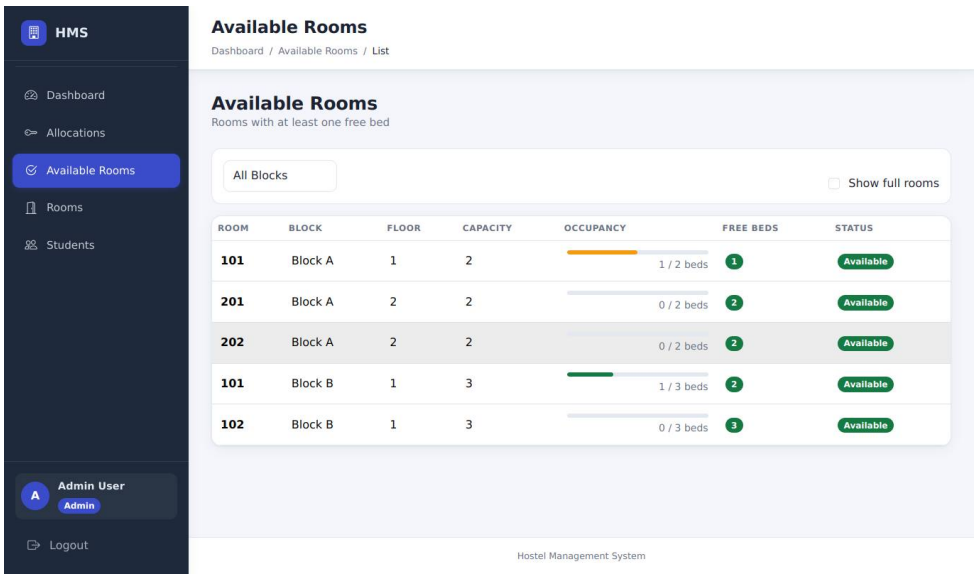
Scenario: Only rooms with free beds are listed

Preconditions: Block A 101 (1/2), Block A 102 (full).

Steps: 1. Sidebar > Available Rooms

Expected Result: Block A 102 (full) is NOT listed; Block A 101 is listed with its free beds.

AVAIL-01_1.jpg — Available Rooms: Block A 102 (full) is not listed
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

40. AVAIL-03 — Block filter and "Show full rooms"

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P2 / UI only

Module: Available rooms

Requirement: RA-08, RA-04

Technique: UI only

Priority: P2

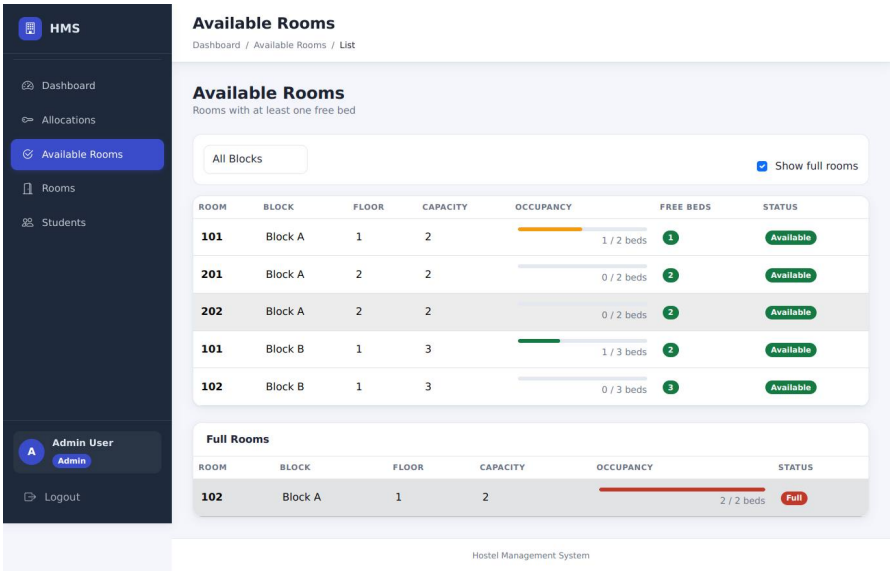
Scenario: Block filter and "Show full rooms"

Preconditions: As AVAIL-01.

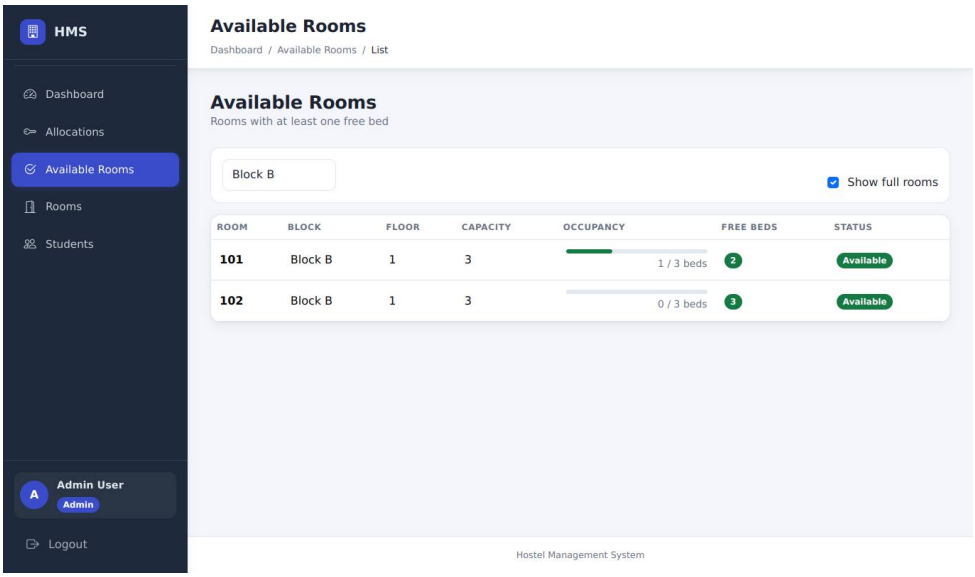
Steps: 1. Tick "Show full rooms"
2. Choose Block B in the block drop-down

Expected Result: Step 1: Block A 102 appears marked full. Step 2: only Block B rooms are shown (no Block A rooms, full or not).

AVAIL-03_1.jpg — Step 1: "Show full rooms" ticked - the full room appears
Reference screenshot of expected state; not verified execution evidence.



AVAIL-03_2.jpg — Step 2: Block B selected - only Block B rooms
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

41. OCC-03 — Full rooms cannot be picked when creating

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P2 / UI only

Module: Occupancy

Requirement: RA-04

Technique: UI only

Priority: P2

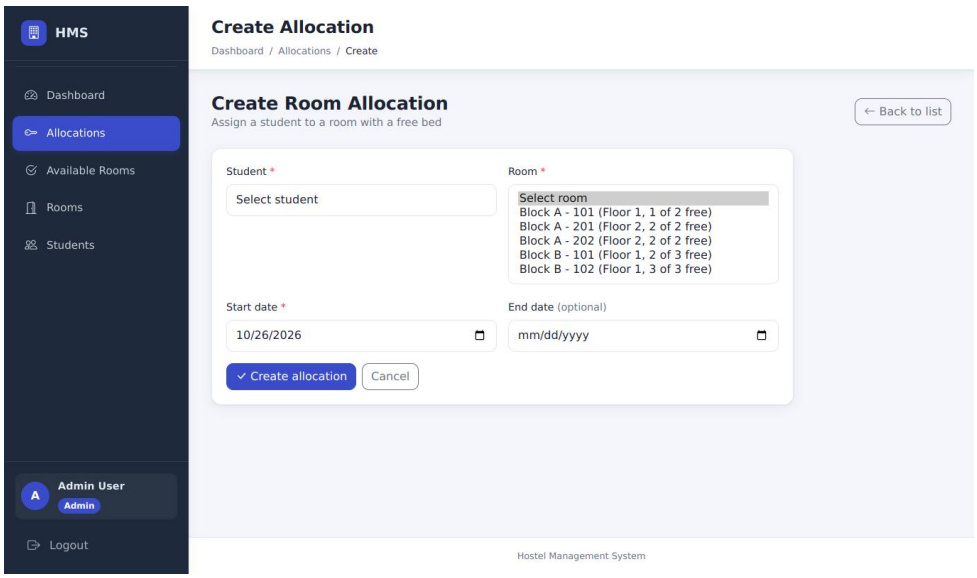
Scenario: Full rooms cannot be picked when creating

Preconditions: Block A 102 is full.

Steps: 1. New Allocation > open the Room drop-down

Expected Result: Block A 102 is not offered.

OCC-03_1.jpg — New Allocation room list (expanded for the screenshot): the full Block A 102 is not offered
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

42. AVAIL-02 — Student can view available rooms (read-only)

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P2 / UI only

Module: Available rooms

Requirement: RA-08

Technique: UI only

Priority: P2

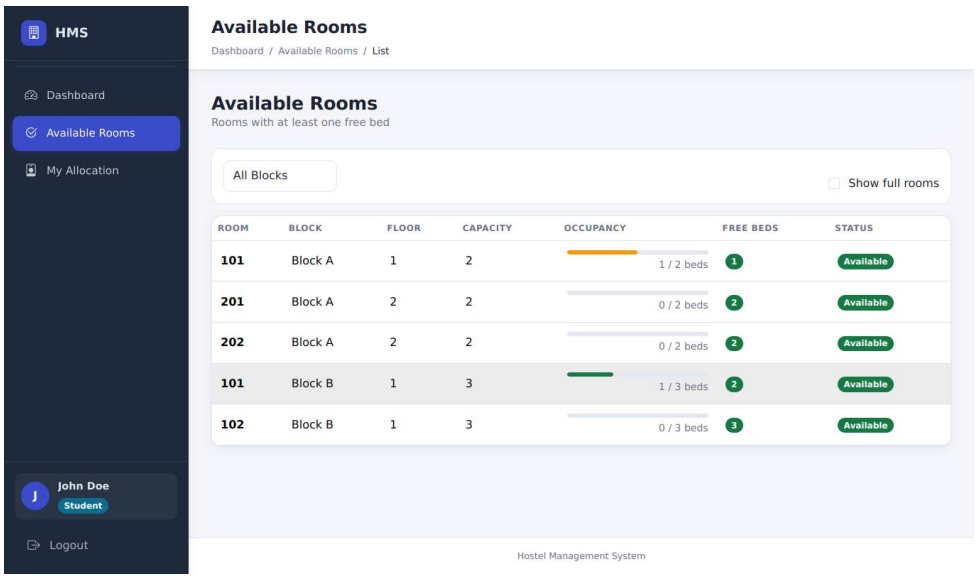
Scenario: Student can view available rooms (read-only)

Preconditions: Logged in as student.

Steps: 1. Sidebar > Available Rooms

Expected Result: Page loads; no allocation actions are offered.

AVAIL-02_1.jpg — Student view of Available Rooms (read-only)
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

43. STUDENT-01 — Student sees own allocation

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P1 / UI only

Module: Student view

Requirement: RA-09

Technique: UI only

Priority: P1

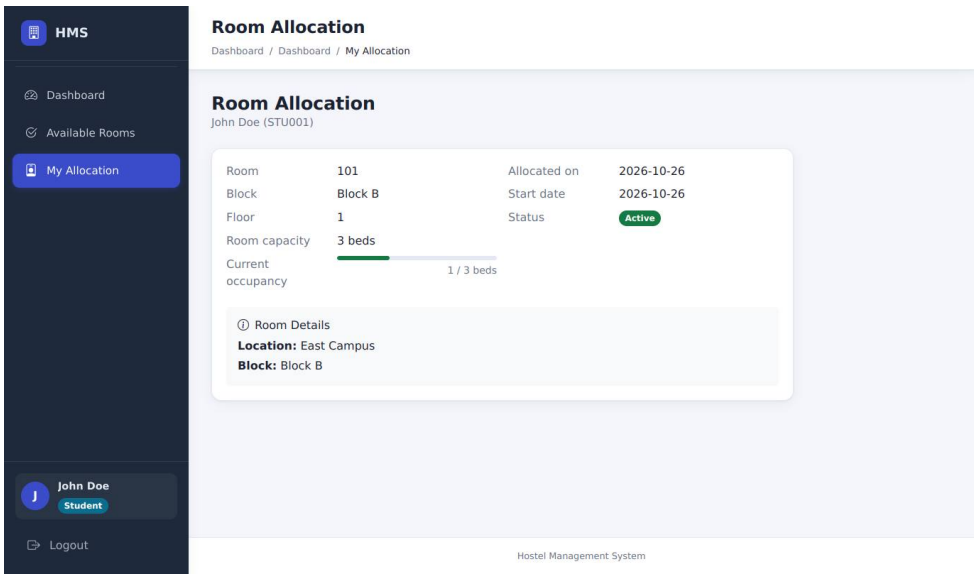
Scenario: Student sees own allocation

Preconditions: Logged in as John Doe (allocated).

Steps: 1. Sidebar > My Allocation

Expected Result: Page shows John Doe's room and block (Block B 101 after REALLOC-01).

STUDENT-01_1.jpg — My Allocation: John Doe's room (Block B 101 after the reallocation)
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

44. STUDENT-02 — Student cannot view another student's allocation

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P1 / UI only

Module: Student view

Requirement: RA-09

Technique: UI only

Priority: P1

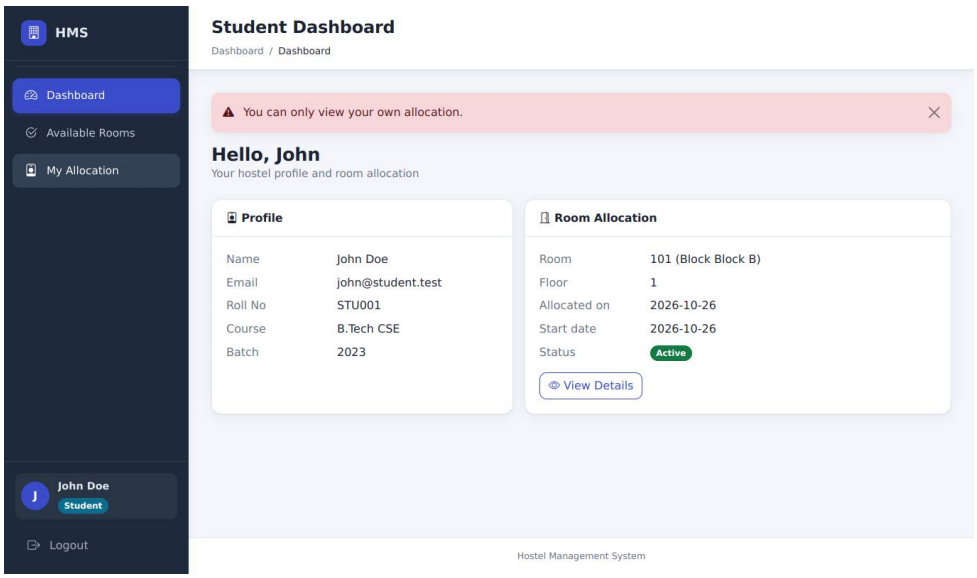
Scenario: Student cannot view another student's allocation

Preconditions: Logged in as John Doe. Jane Smith has student id 2.

Steps: 1. In the address bar open http://127.0.0.1:5000/allocation/student/2

Expected Result: Redirected to the dashboard with "You can only view your own allocation."

STUDENT-02_1.jpg — Opening another student's allocation: redirected to the dashboard with a message
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

45. AUTHZ-01 — Admin reaches all management pages

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P1 / UI only

Module: Authorization

Requirement: RA-10

Technique: UI only

Priority: P1

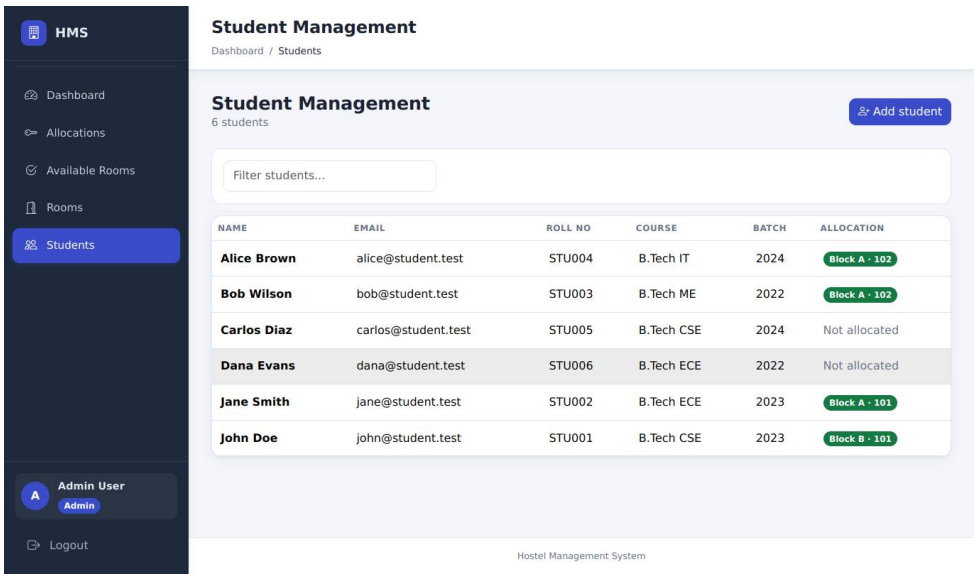
Scenario: Admin reaches all management pages

Preconditions: Logged in as admin.

Steps: 1. Open Allocations, New Allocation, Available Rooms, Rooms, Students from the sidebar/buttons

Expected Result: All pages load without a permission error.

AUTHZ-01_1.jpg — Admin can open the Students page (all management pages load)
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

46. AUTHZ-02 — Warden reaches all management pages

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P2 / UI only

Module: Authorization

Requirement: RA-10

Technique: UI only

Priority: P2

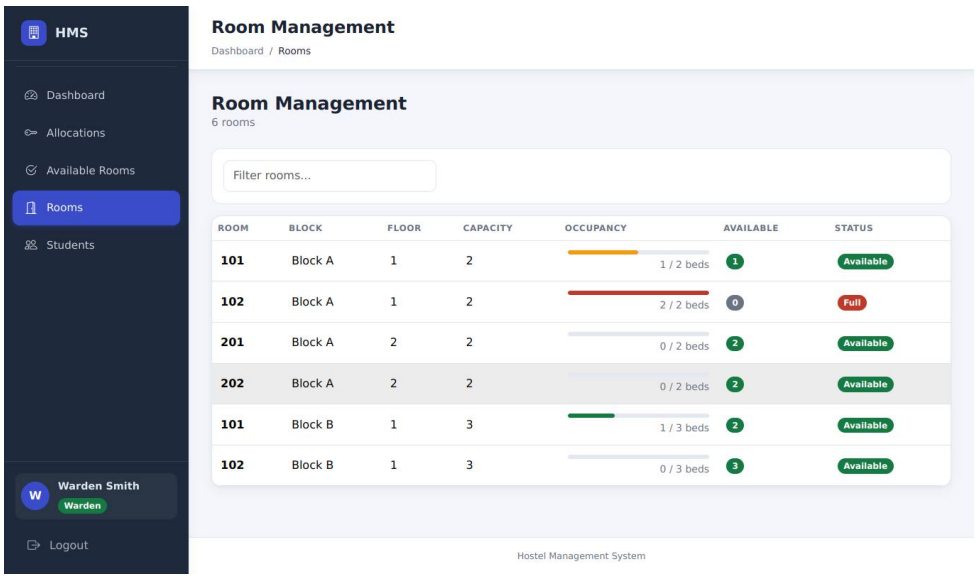
Scenario: Warden reaches all management pages

Preconditions: Logged in as warden.

Steps: 1. Repeat the AUTHZ-01 pages

Expected Result: All pages load.

AUTHZ-02_1.jpg — Warden can open the Rooms page
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

47. AUTHZ-03 — Student blocked from allocation management

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P1 / UI only

Module: Authorization

Requirement: RA-10

Technique: UI only

Priority: P1

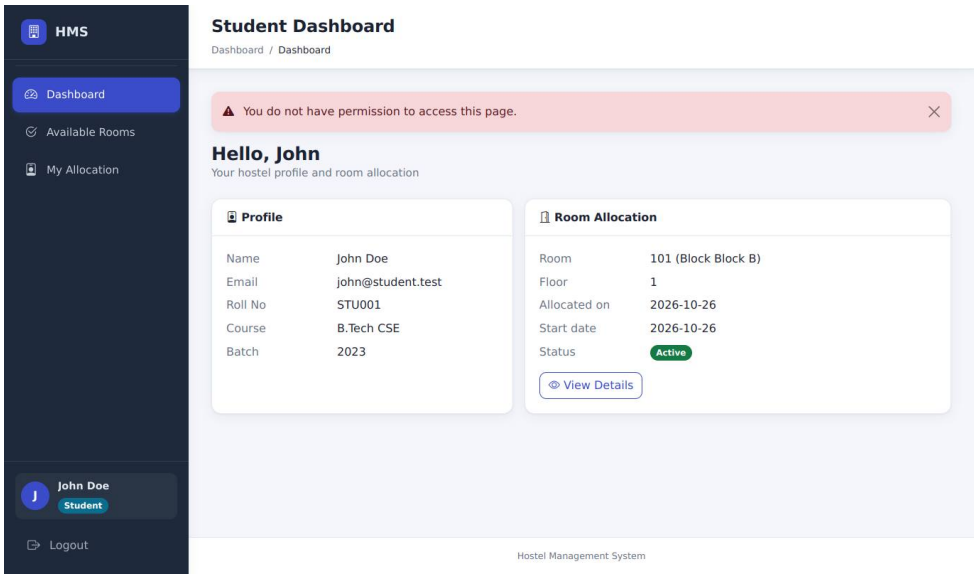
Scenario: Student blocked from allocation management

Preconditions: Logged in as student.

Steps: 1. Open http://127.0.0.1:5000/allocation/
2. Open /allocation/create

Expected Result: Each redirects to the dashboard with "You do not have permission to access this page."

AUTHZ-03_1.jpg — Student opening /allocation/: redirected with "You do not have permission..."
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

48. AUTHZ-05 — Accountant blocked from allocations

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P2 / UI only

Module: Authorization

Requirement: RA-10

Technique: UI only

Priority: P2

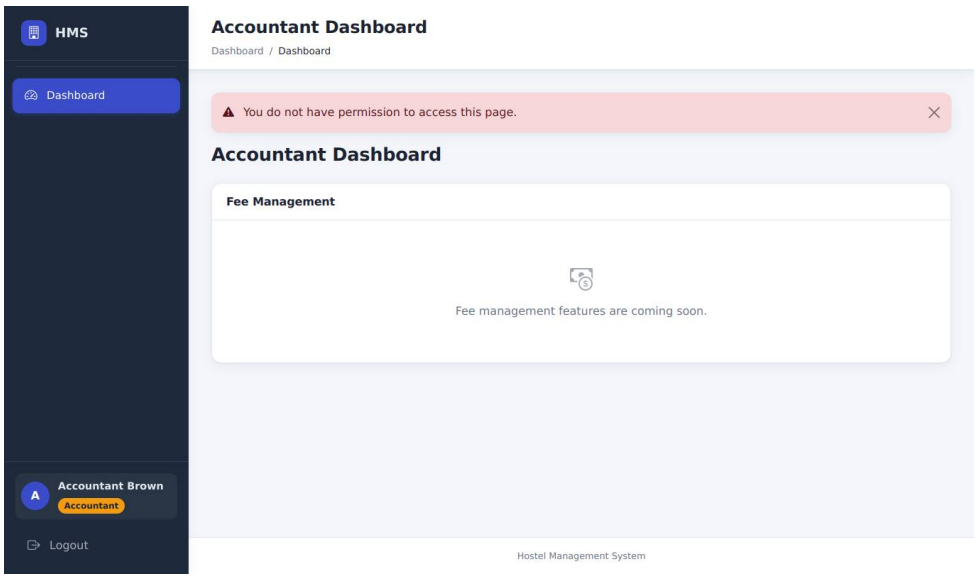
Scenario: Accountant blocked from allocations

Preconditions: Logged in as accountant@hms.test / accountant123.

Steps: 1. Open /allocation/

Expected Result: Redirected with "You do not have permission to access this page."

AUTHZ-05_1.jpg — Accountant opening /allocation/: permission message
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

49. AUTHZ-07 — Student blocked from rooms and students pages

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P1 / UI only

Module: Authorization

Requirement: RA-10

Technique: UI only

Priority: P1

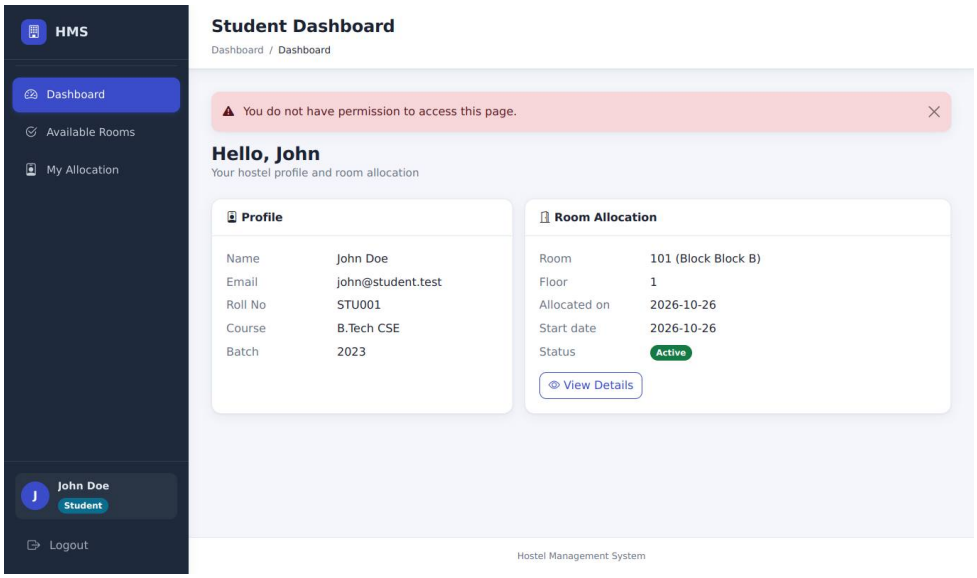
Scenario: Student blocked from rooms and students pages

Preconditions: Logged in as student.

Steps: 1. Open /rooms
2. Open /students

Expected Result: Both redirect with "You do not have permission to access this page."

AUTHZ-07_1.jpg — Student opening /rooms: permission message
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

50. AUTHZ-10 — Student cannot open another allocation's reallocate page

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P2 / UI only

Module: Authorization

Requirement: RA-10

Technique: UI only

Priority: P2

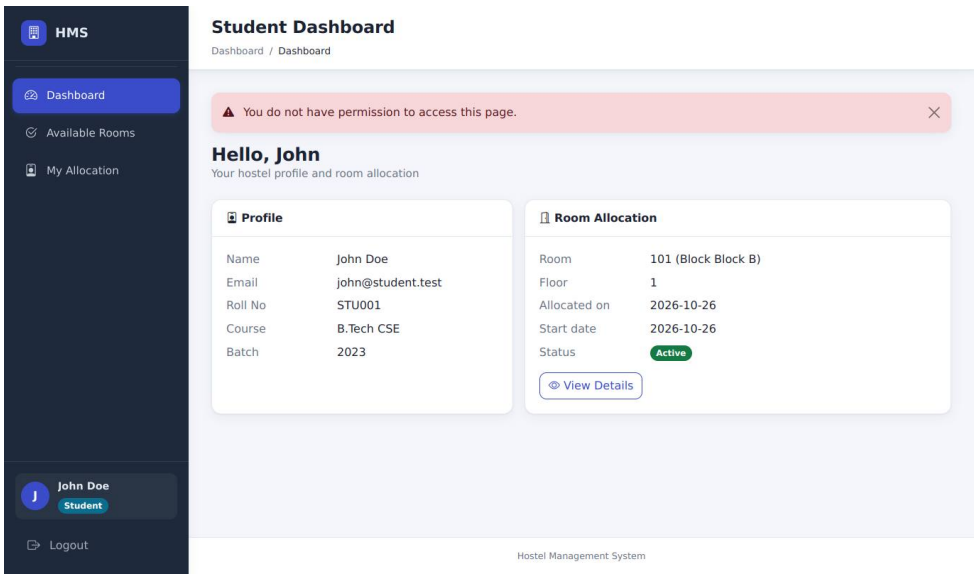
Scenario: Student cannot open another allocation's reallocate page

Preconditions: Logged in as student. An allocation with id 2 exists.

Steps: 1. Open http://127.0.0.1:5000/allocation/2/reallocate

Expected Result: Redirected with "You do not have permission to access this page."

AUTHZ-10_1.jpg — Student opening /allocation/2/reallocate: permission message
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

51. AUTHZ-11 — Student cannot cancel an allocation

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P2 / UI + browser DevTools

Module: Authorization

Requirement: RA-10, RA-12

Technique: UI + browser DevTools

Priority: P2

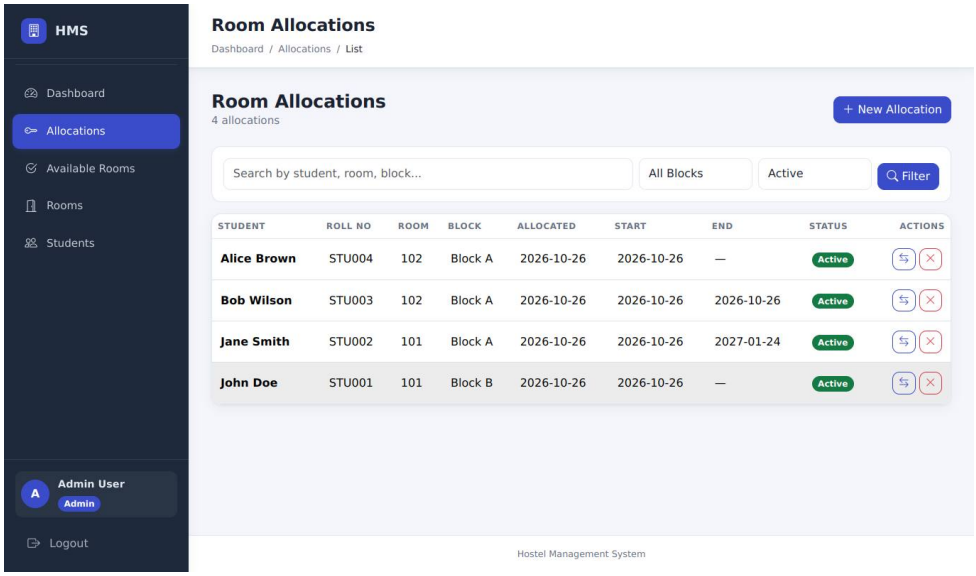
Scenario: Student cannot cancel an allocation

Preconditions: Logged in as student. Allocation id 2 is Active.

Steps: 1. Open DevTools > Console and run:
fetch("/allocation/2/cancel",{method:"POST",headers:{"Content-Type":"application/x-www-form-urlencoded"},body:"csrf_token="+document.querySelector("input[name=csrf_token]").value})
2. Log in as admin and check allocation 2

Expected Result: The student request is refused (redirect to the dashboard / permission message); allocation 2 is still Active.

AUTHZ-11_1.jpg — Admin check after the student's forged request: Jane's allocation (id 2) is still Active
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

52. SEC-01 — SQL injection in the search box

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P1 / UI only

Module: Security

Requirement: RA-14

Technique: UI only

Priority: P1

Scenario: SQL injection in the search box

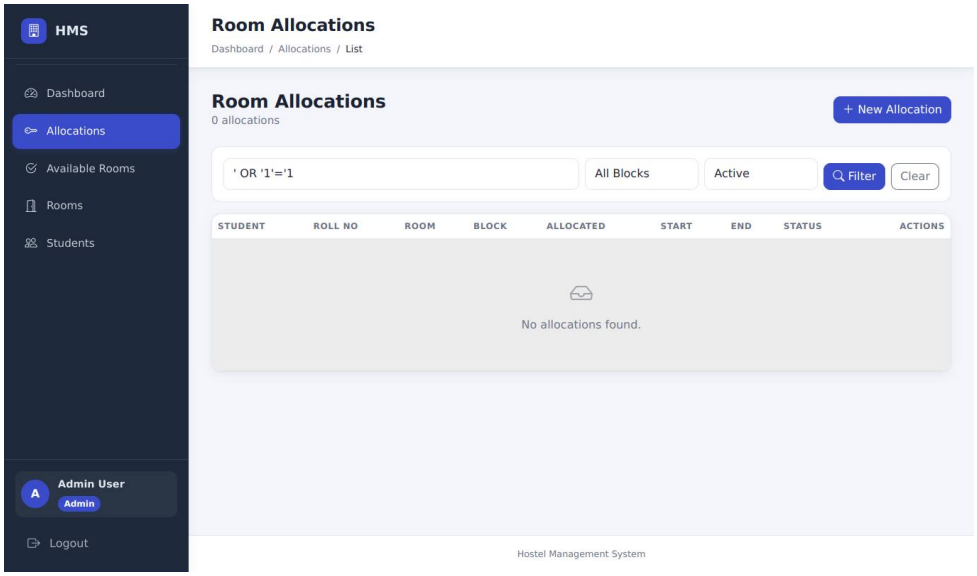
Preconditions: Logged in as admin.

Steps: 1. Allocations list > search for: ' OR '1'='1
2. Search for: '; DROP TABLE rooms; --

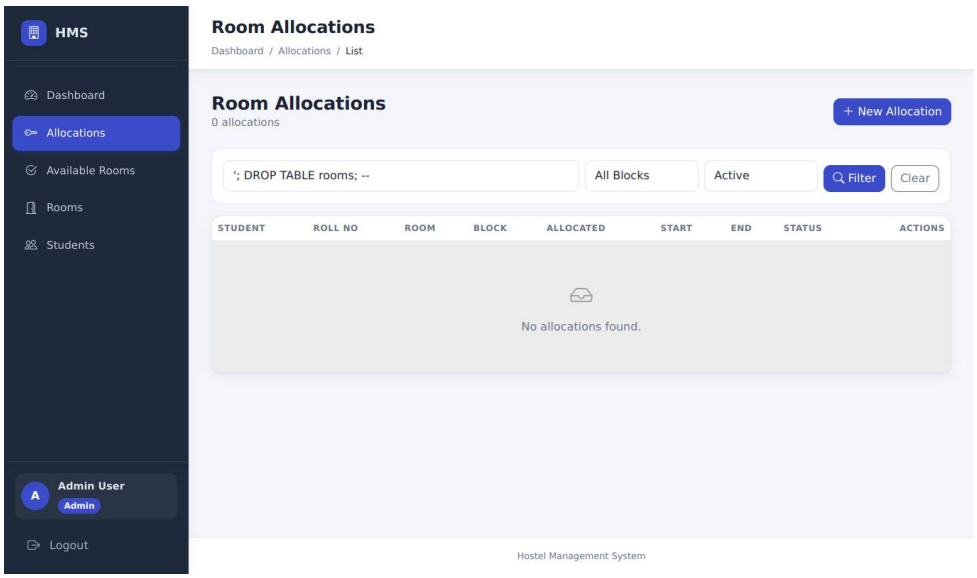
Input Data: SQL payloads

Expected Result: Page renders normally ("No allocations found." or a normal list); no error page; Rooms page still shows all rooms.

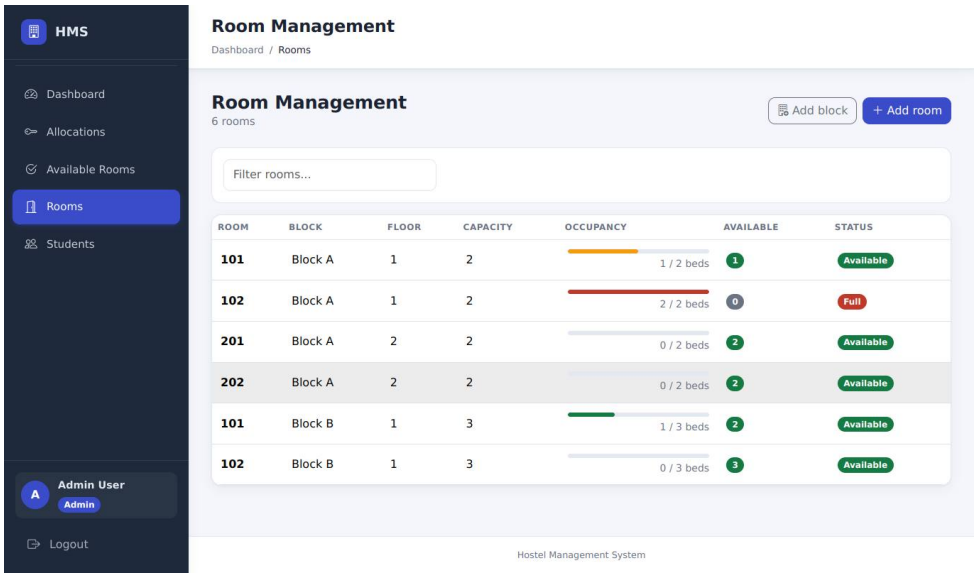
SEC-01_1.jpg — Payload 1 in the search box: normal page, no error
Reference screenshot of expected state; not verified execution evidence.



SEC-01_2.jpg — Payload 2: normal page; the Rooms table is unaffected
Reference screenshot of expected state; not verified execution evidence.



SEC-01_3.jpg — Rooms page afterwards: all rooms are still there
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

53. SEC-03 — XSS in the search box

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P1 / UI only

Module: Security

Requirement: RA-14

Technique: UI only

Priority: P1

Scenario: XSS in the search box

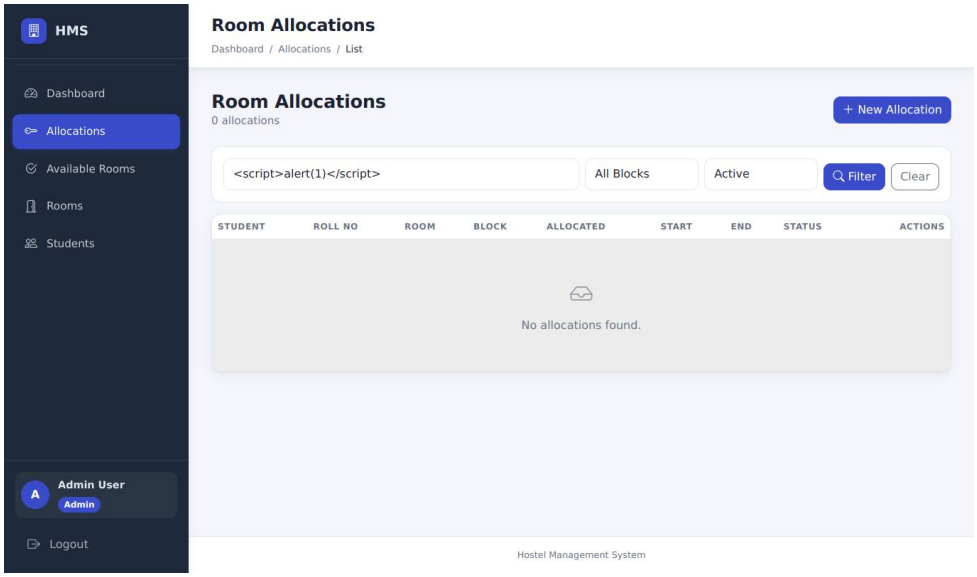
Preconditions: Logged in as admin.

Steps: 1. Allocations list > search for: <script>alert(1)</script>

Input Data: <script>alert(1)</script>

Expected Result: No alert box appears; the text is displayed as plain text in the search field (escaped).

SEC-03_1.jpg — XSS payload: shown as plain text in the search field, no alert box
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

54. SEC-05 — Forged CSRF token is rejected

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P2 / UI + browser DevTools

Module: Security

Requirement: RA-12

Technique: UI + browser DevTools

Priority: P2

Scenario: Forged CSRF token is rejected

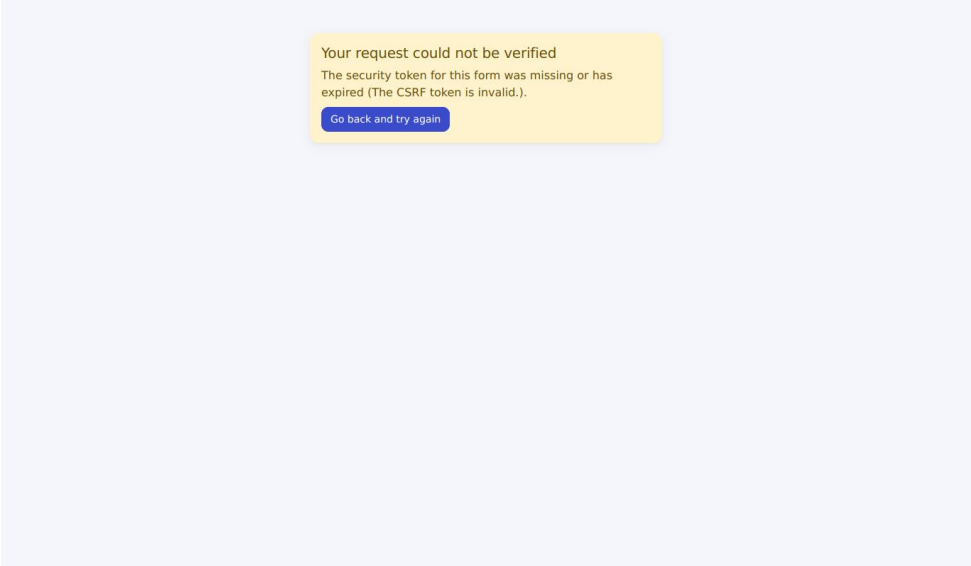
Preconditions: Logged in as admin.

Steps: 1. Open New Allocation, fill the form
2. In DevTools set the hidden csrf_token value to "forged"
3. Submit

Input Data: csrf_token=forged

Expected Result: Page "Your request could not be verified" (HTTP 400); nothing created.

SEC-05_1.jpg — Forged CSRF token: HTTP 400 "Your request could not be verified"
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

55. SEC-06 — No session bleed between users

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P2 / UI only

Module: Security

Requirement: RA-11

Technique: UI only

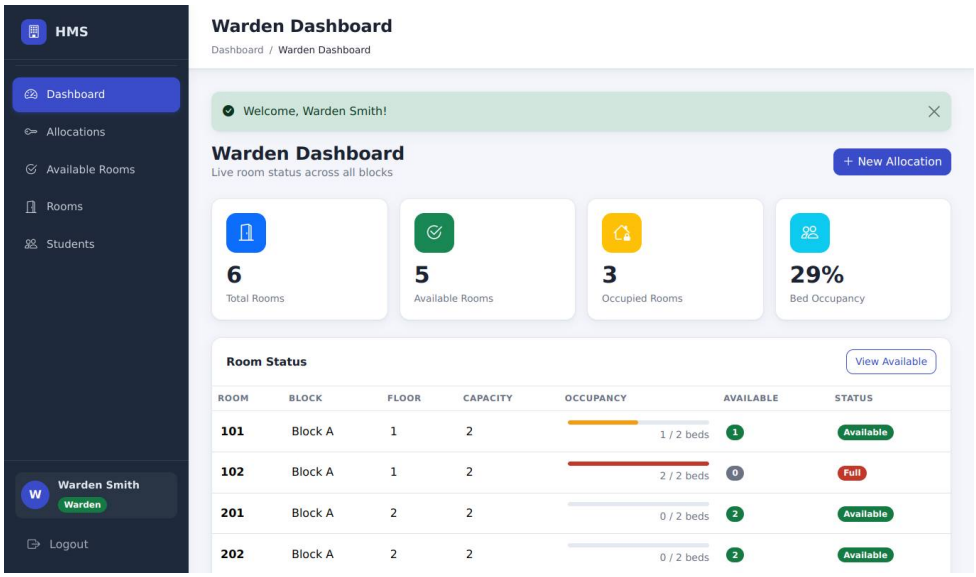
Priority: P2

Scenario: No session bleed between users

Steps: 1. Log in as admin, then Logout
2. Log in as warden

Expected Result: Warden Dashboard shows the warden's name and no admin-only content.

SEC-06_1.jpg — After admin logout and warden login: the Warden dashboard only
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

56. SEC-07 — Demo credentials are not displayed

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P1 / UI only

Module: Security

Requirement: RA-13

Technique: UI only

Priority: P1

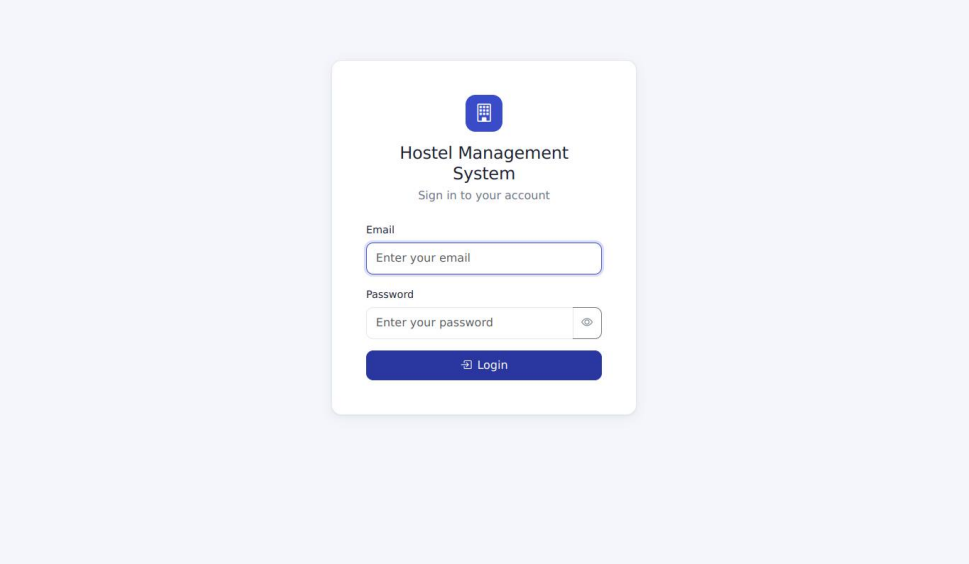
Scenario: Demo credentials are not displayed

Preconditions: Server started normally (SHOW_DEMO_CREDENTIALS not set).

Steps: 1. Open the login page and read it completely

Expected Result: No demo accounts or passwords are listed on the page.

SEC-07_1.jpg — Login page: no demo accounts or passwords are displayed
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

57. DASH-01 — Dashboards match the data

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P2 / UI only

Module: Dashboards

Requirement: RA-15

Technique: UI only

Priority: P2

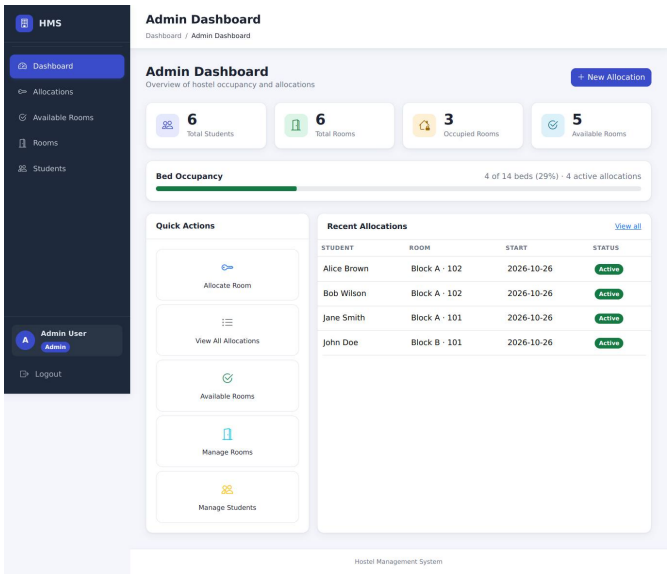
Scenario: Dashboards match the data

Preconditions: Allocations from the cases above exist.

Steps: 1. View the Admin, Warden and Student dashboards in turn

Expected Result: Admin: Total Students = 6 and Total Rooms = 6 equal what the Students and Rooms pages show; active allocations and occupancy bar agree with the Allocations list. Student: shows own room.

DASH-01_1.jpg — Admin dashboard with the data created by the previous cases
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

58. UI-01 — Mobile layout (390 px wide)

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P1 / UI + browser DevTools

Module: UI

Requirement: UI-02

Technique: UI + browser DevTools

Priority: P1

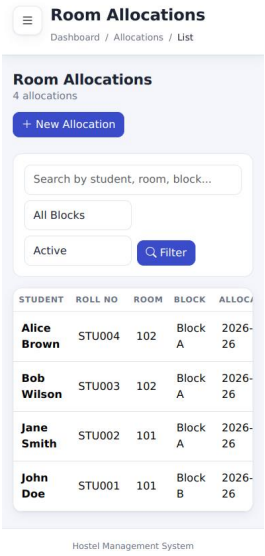
Scenario: Mobile layout (390 px wide)

Preconditions: Logged in as admin; DevTools device toolbar set to 390 x 844.

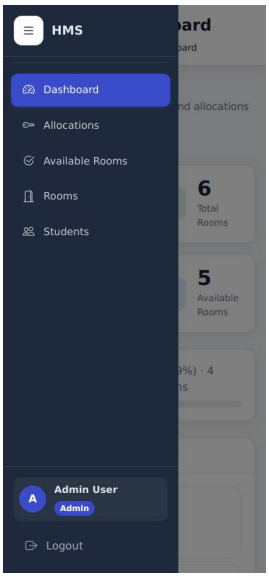
Steps: 1. Visit Dashboard, Allocations, Available Rooms, Rooms, Students, New Allocation
2. On each page: scroll horizontally; look at the title next to the hamburger button
3. Tap the hamburger button

Expected Result: No page scrolls sideways (wide tables scroll inside their card); the hamburger never covers the page title; tapping it opens the navigation drawer.

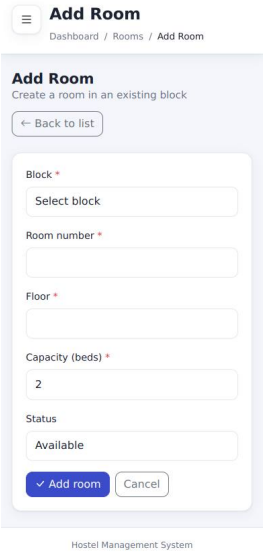
UI-01_1.jpg — Mobile (390 px): Allocations - no sideways scrolling, hamburger beside the title
Reference screenshot of expected state; not verified execution evidence.



UI-01_2.jpg — Mobile: the navigation drawer after tapping the hamburger button
Reference screenshot of expected state; not verified execution evidence.



UI-01_3.jpg — Mobile: a form page (Add room) also fits the screen
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

59. UI-02 — Menus match the role

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P2 / UI only

Module: UI

Requirement: UI-01

Technique: UI only

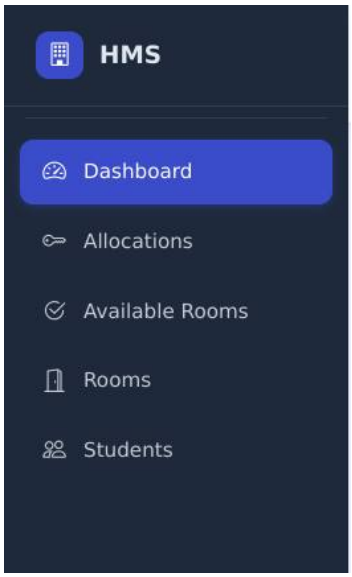
Priority: P2

Scenario: Menus match the role

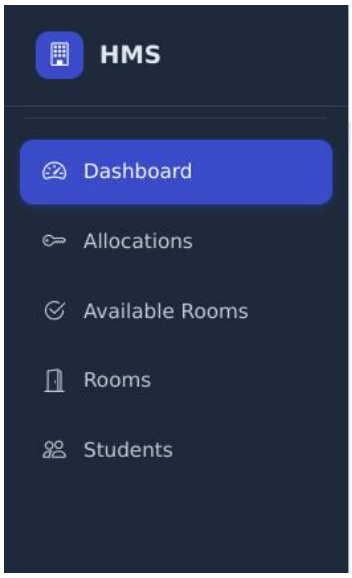
Steps: 1. Log in as admin, then warden, then student; read the sidebar each time

Expected Result: Admin and warden: Dashboard, Allocations, Available Rooms, Rooms, Students. Student: Dashboard, Available Rooms, My Allocation. "My Allocation" never appears for admin/warden.

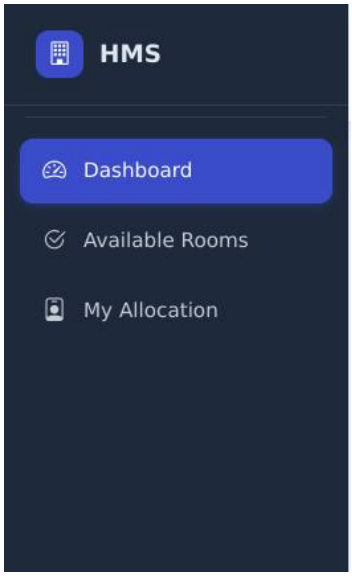
UI-02_1.jpg — Admin sidebar
Reference screenshot of expected state; not verified execution evidence.



UI-02_2.jpg — Warden sidebar
Reference screenshot of expected state; not verified execution evidence.



UI-02_3.jpg — Student sidebar
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

60. UI-03 — Single logout; cache not reused after logout

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P1 / UI only

Module: UI

Requirement: UI-05, RA-11

Technique: UI only

Priority: P1

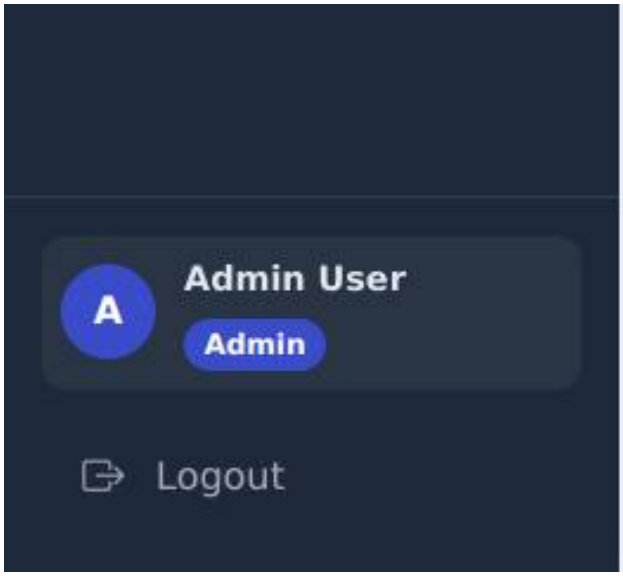
Scenario: Single logout; cache not reused after logout

Preconditions: Logged in as admin.

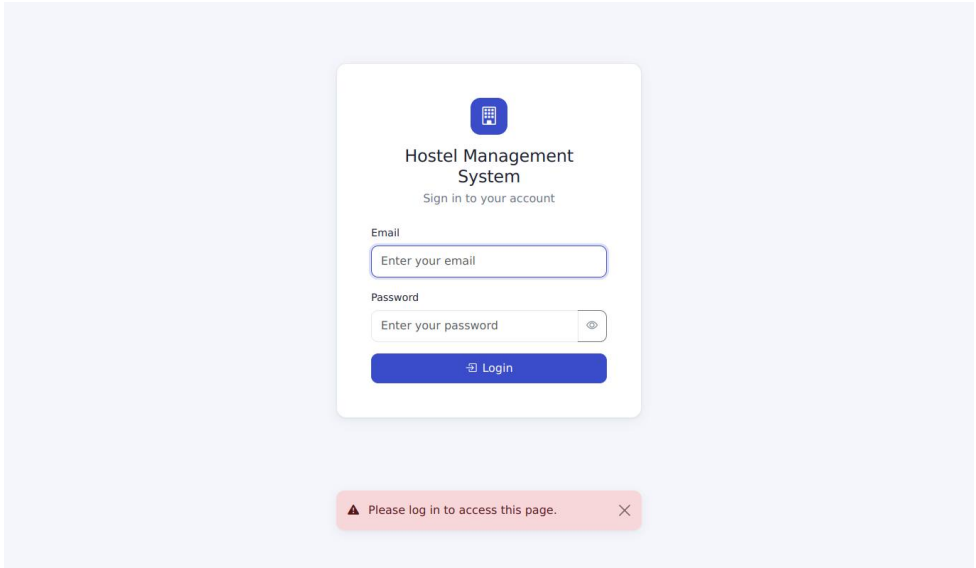
Steps: 1. Count the Logout controls visible on screen
2. Click Logout
3. Press the browser Back button

Expected Result: Exactly one Logout control. After Back the dashboard content is NOT shown; you are sent to the login page.

UI-03_1.jpg — Exactly one Logout control (bottom of the sidebar)
Reference screenshot of expected state; not verified execution evidence.



UI-03_2.jpg — After Logout and the browser Back button: the login page, not the dashboard
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

61. ADD-01 — Add a hostel block

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P2 / UI only

Module: Data entry

Requirement: RA-16

Technique: UI only

Priority: P2

Scenario: Add a hostel block

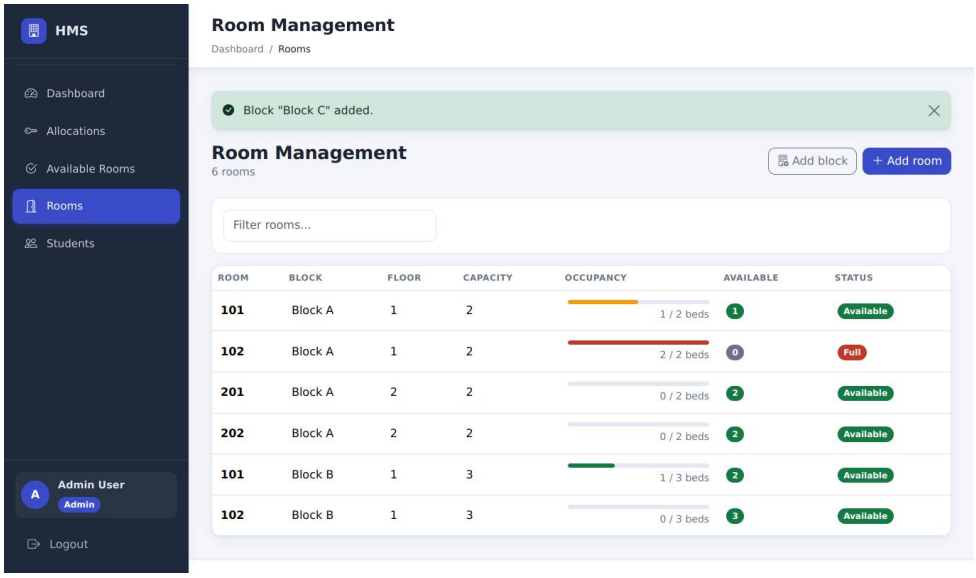
Preconditions: Logged in as admin. Run AFTER all other cases (it changes the totals checked by DASH-01).

Steps: 1. Sidebar > Rooms > Add block
2. Block name: Block C; Location: South Campus
3. Click Add block

Input Data: Block C / South Campus

Expected Result: Redirected to the Rooms page with the message "Block "Block C" added."; Block C appears in the block filter of Allocations.

ADD-01_1.jpg — Block C added: message on the Rooms page
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

62. ADD-02 — Add a room

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P2 / UI only

Module: Data entry

Requirement: RA-16

Technique: UI only

Priority: P2

Scenario: Add a room

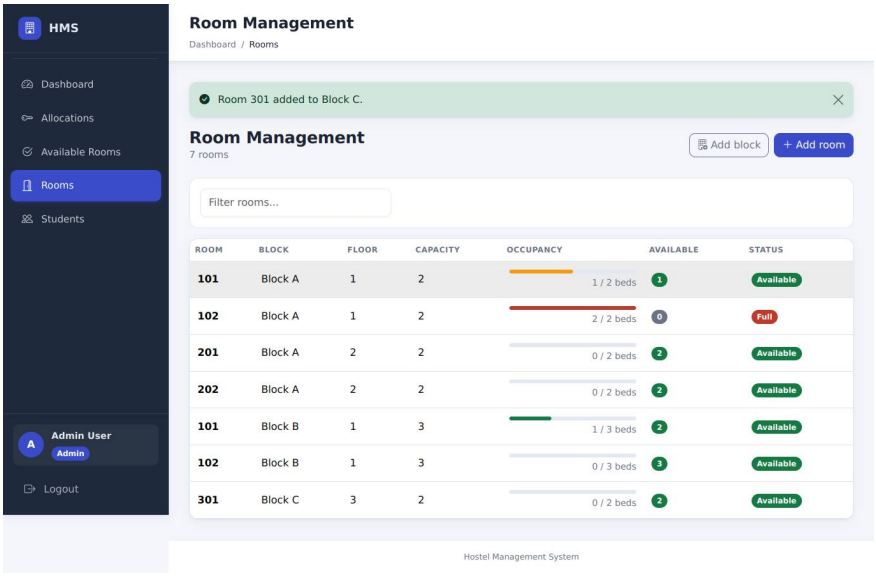
Preconditions: ADD-01 done.

Steps: 1. Rooms > Add room
2. Block: Block C; Room number: 301; Floor: 3; Capacity: 2; Status: Available
3. Click Add room

Input Data: Block C / 301 / floor 3 / 2 beds

Expected Result: Message "Room 301 added to Block C."; the Rooms table shows Block C 301 with 0 / 2 beds" and status Available; the New Allocation room list offers it."

ADD-02_1.jpg — Room 301 added to Block C: message and the new row in the table
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

63. ADD-03 — Add a student and log in as that student

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P2 / UI only

Module: Data entry

Requirement: RA-16

Technique: UI only

Priority: P2

Scenario: Add a student and log in as that student

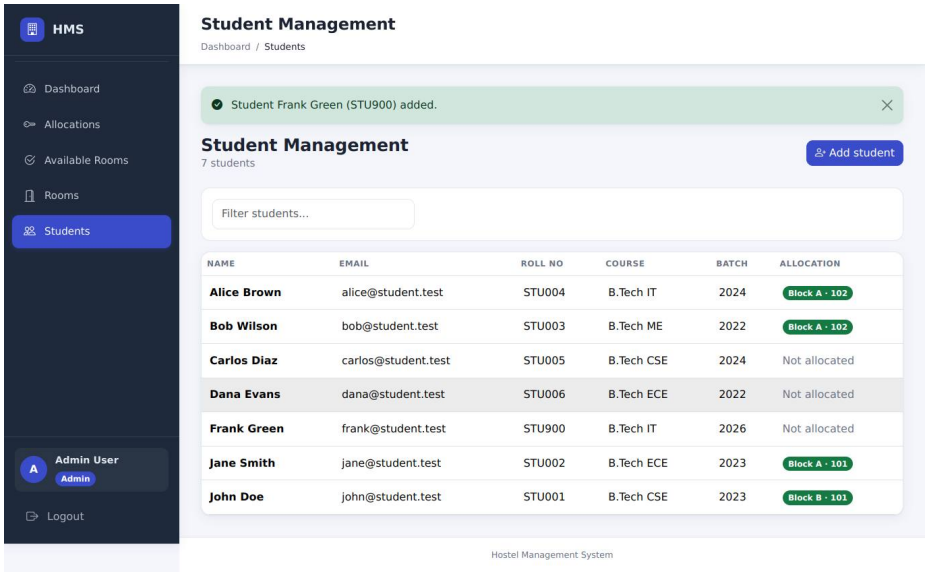
Preconditions: ADD-02 done.

Steps: 1. Students > Add student
2. Name: Frank Green; E-mail: frank@student.test; Roll number: STU900; Course: B.Tech IT; Batch: 2026; Password and confirmation: Frank#Pass123
3. Click Add student
4. Log out; log in as frank@student.test / Frank#Pass123

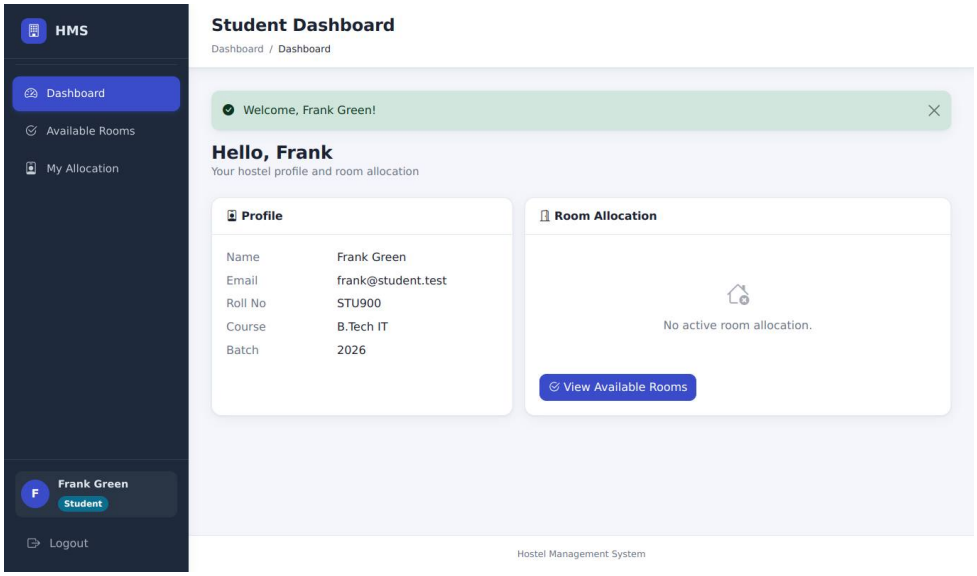
Input Data: Frank Green / STU900

Expected Result: Message "Student Frank Green (STU900) added."; Frank appears in the Students list with no allocation; the new login works and shows the student menu (Dashboard, Available Rooms, My Allocation).

ADD-03_1.jpg — Step 3: Frank Green added - message and the new row in the Students list
Reference screenshot of expected state; not verified execution evidence.



ADD-03_2.jpg — Step 4: logging in as Frank works and shows the student menu
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

64. ADD-04 — Validation and duplicates on the add forms

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P2 / UI only

Module: Data entry

Requirement: RA-16

Technique: UI only

Priority: P2

Scenario: Validation and duplicates on the add forms

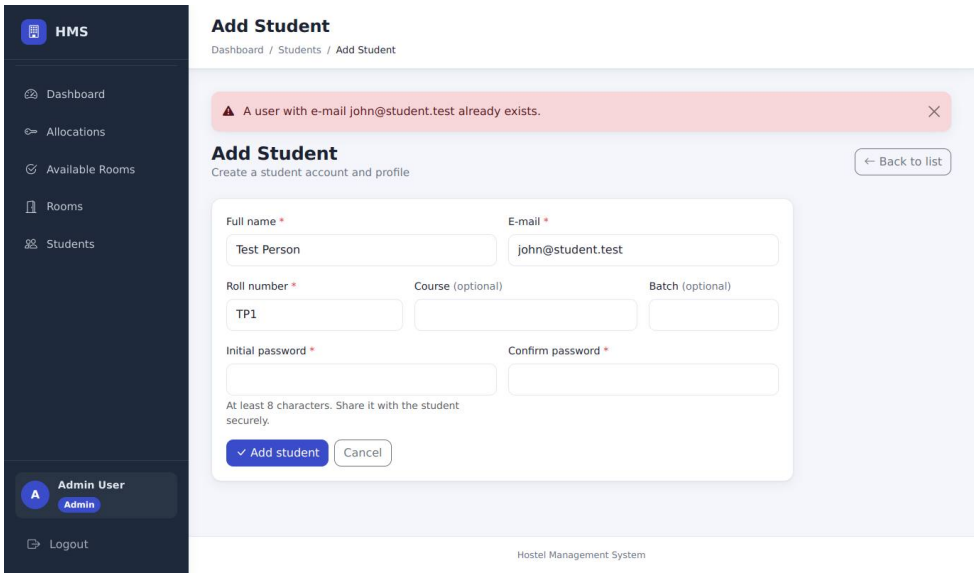
Preconditions: Logged in as admin; ADD-03 done.

Steps: 1. Add student with E-mail john@student.test and everything else valid
2. Add student with Roll number STU900 and a new e-mail
3. Add student with Password abc and confirmation abc
4. Add room: Block C, room number 301 again
5. Add block named "block c"

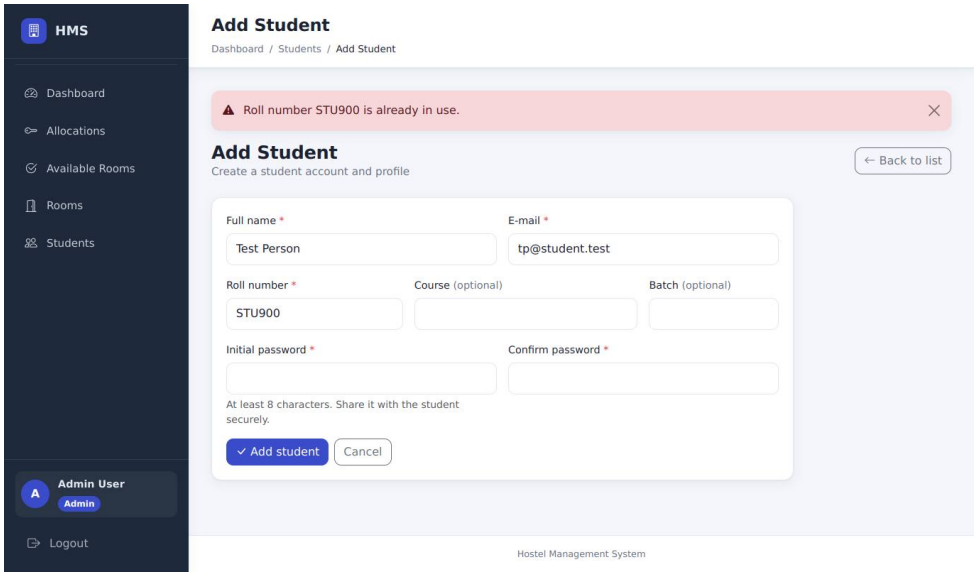
Input Data: see steps

Expected Result: 1: "A user with e-mail john@student.test already exists."; 2: "Roll number STU900 is already in use."; 3: "Password must be 8 to 128 characters long."; 4: "Room 301 already exists in Block C."; 5: "A block named "block c" already exists." Every form is shown again with your entries kept (passwords are never refilled); nothing is created.

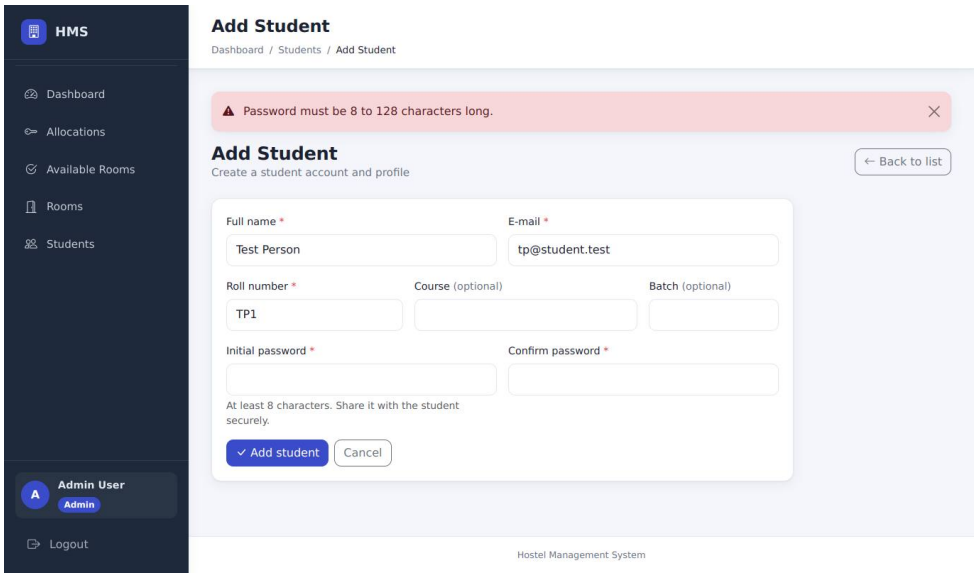
ADD-04_1.jpg — 1: duplicate e-mail rejected
Reference screenshot of expected state; not verified execution evidence.



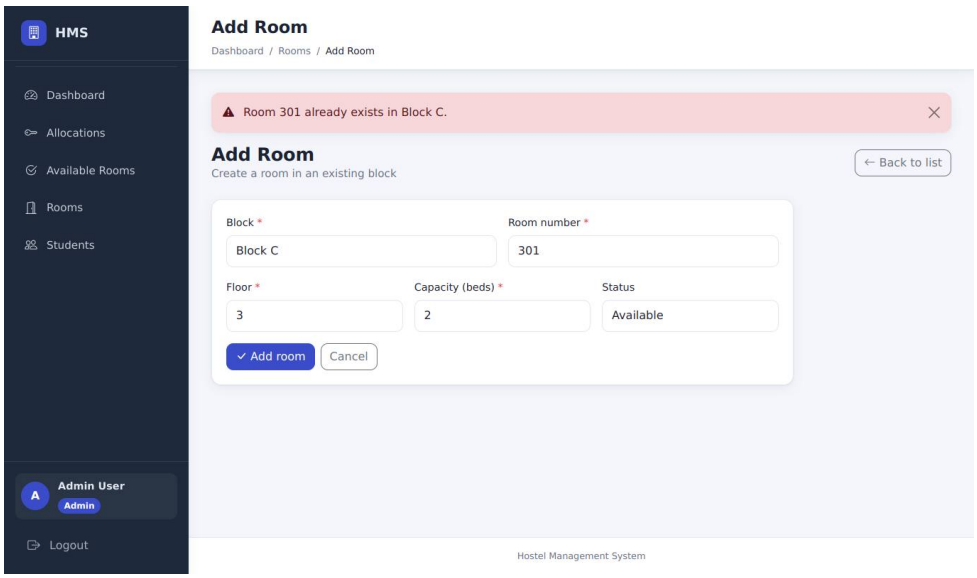
ADD-04_2.jpg — 2: duplicate roll number rejected
Reference screenshot of expected state; not verified execution evidence.



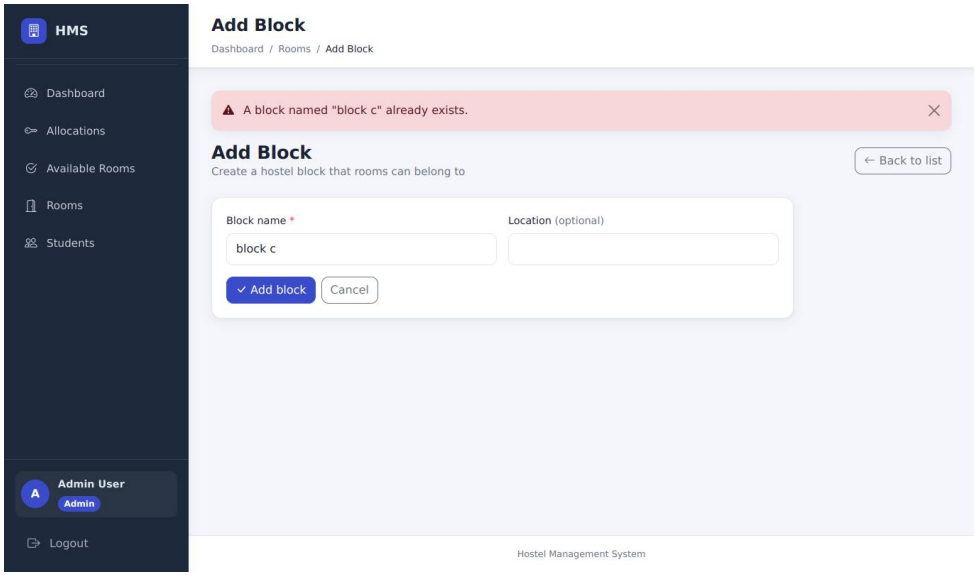
ADD-04_3.jpg — 3: too-short password rejected
Reference screenshot of expected state; not verified execution evidence.



ADD-04_4.jpg — 4: duplicate room number in the same block rejected
Reference screenshot of expected state; not verified execution evidence.



ADD-04_5.jpg — 5: duplicate block name rejected (case-insensitive)
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.

65. ADD-05 — Only admins can add data

Execution	Executed (Y) — outcome not independently verified
Priority / Technique	P2 / UI only

Module: Data entry

Requirement: RA-16, RA-10

Technique: UI only

Priority: P2

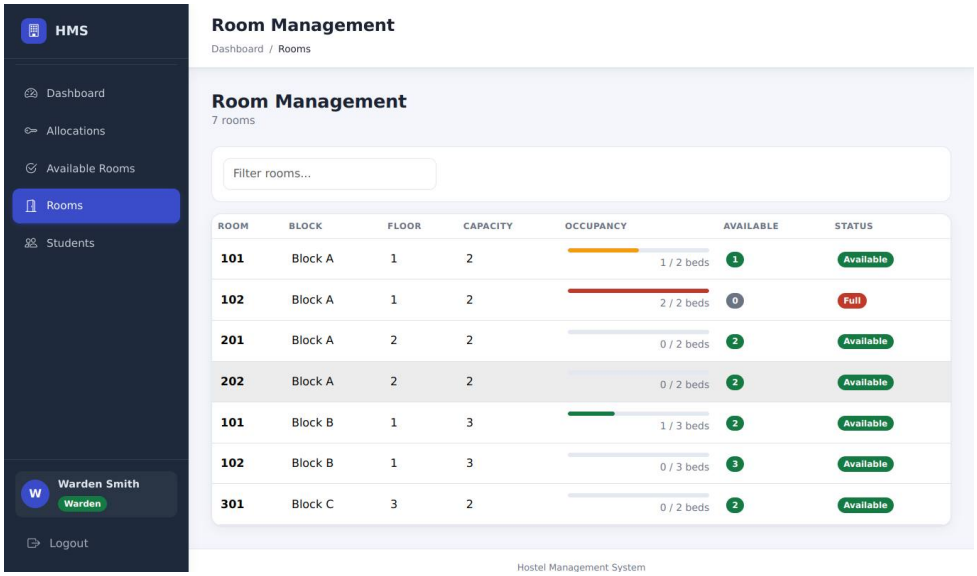
Scenario: Only admins can add data

Preconditions: Logged in as warden, then as student.

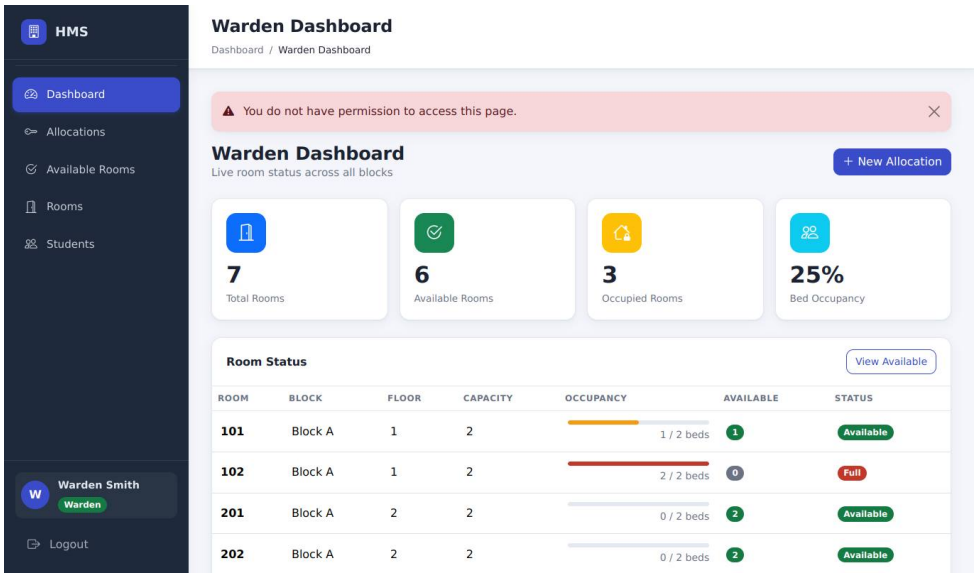
Steps: 1. As warden open Rooms and Students: look for Add buttons
2. As warden open http://127.0.0.1:5000/rooms/new
3. As student open http://127.0.0.1:5000/students/new

Expected Result: Warden sees no Add buttons; both URLs redirect to the dashboard with "You do not have permission to access this page."

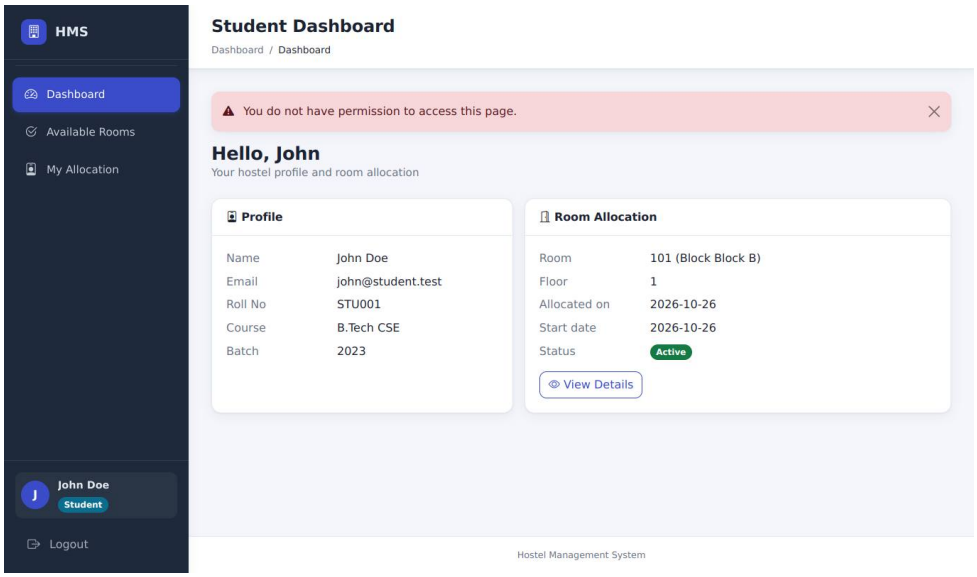
ADD-05_1.jpg — Warden: the Rooms page has no Add buttons
Reference screenshot of expected state; not verified execution evidence.



ADD-05_2.jpg — Warden opening /rooms/new: permission message
Reference screenshot of expected state; not verified execution evidence.



ADD-05_3.jpg — Student opening /students/new: permission message
Reference screenshot of expected state; not verified execution evidence.



Actual result recorded: Reference screenshot(s) supplied; live application behavior not independently verified.